

NOTTINGHAM TRENT UNIVERSITY
SCHOOL OF SCIENCE AND TECHNOLOGY

**Personalized Gait and TUG-Related
Mobility Analysis Using Wearable IMU
Sensors to Support Rehabilitation
Monitoring**

by

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"Full Name of the Author"

in

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**Project report in part fulfilment
of the requirements for the degree of
Master of Science**

In
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ABSTRACT

Functional mobility assessments such as the Timed Up and Go (TUG) test are widely used to evaluate gait, balance and movement performance. However, conventional stopwatch-based assessment reduces a complex sequence of movements to a single completion time. This project developed a wearable Inertial Measurement Unit (IMU) system for personalised gait analysis and instrumented TUG assessment, with the aim of supporting objective rehabilitation monitoring through individual movement characteristics.

Three wearable sensing nodes, positioned at the lumbar region, thigh and knee, were used to collect tri-axial acceleration and angular-velocity signals at 50 Hz. Labelled movement data were collected for sitting, standing, walking and TUG-related activity. The signals were filtered, aligned across the three sensor nodes and segmented into two-second windows with 50% overlap. Four activity-recognition approaches were investigated: Random Forest, one-dimensional Convolutional Neural Network (1D-CNN), Long Short-Term Memory (LSTM), and Conv-LSTM. Generalisation was evaluated using Leave-One-Subject-Out (LOSO) cross-validation so that participants used for testing were not represented during model training.

Random Forest was selected as the most suitable model for the prototype because it provided strong performance on unseen participants, useful feature-level interpretation and sufficiently low computational cost. Across 200 benchmark runs, the tuned Random Forest achieved a median single-window inference time of approximately 44.9 ms, a mean of 73.6 ms, and a 95th-percentile latency of approximately 59.5 ms. These values remained well below the nominal one-second interval between successive overlapping windows, supporting near-real-time operation. The selected model was integrated into a live monitoring application capable of recognising sitting, standing and walking while deriving additional mobility measures, including estimated step count, cadence, elapsed activity time, and sit-to-stand, stand-to-sit, stand-to-walk and walk-to-stand transitions.

The study demonstrates that multi-location wearable IMUs, participant-independent activity recognition, and interpretable mobility measurements can be combined within a proof-of-concept assessment tool. The system is not presented as a clinically validated fall-prediction model. Instead, it provides objective mobility measurements that may support longitudinal rehabilitation monitoring and investigation of gait characteristics associated with functional decline and fall risk.

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CHAPTER 1

INTRODUCTION

Introduction

Falls among older adults represent a substantial public-health challenge. They contribute to approximately 255,000 falls-related emergency hospital admissions each year among people aged 65 and over in England and place considerable pressure on health and social-care services. Falls have also commonly been estimated to cost the NHS more than £2.3 billion annually, although this estimate is based on older cost data and should therefore be interpreted cautiously (Macchiavello et al., 2025; Whitney, 2026).

Conventional functional mobility assessments, including the Timed Up and Go (TUG) test, provide clinically useful measures of mobility. However, they are generally administered at discrete assessment points and, in the case of TUG, reduce a multi-stage movement task to one overall completion time (Podsiadlo and Richardson, 1991). Systematic reviews also suggest that these assessments should not be used in isolation for fall-risk prediction, while sensor-based approaches may provide additional objective information about movement patterns that are not captured by total task duration alone (Barry et al., 2014; Beck Jepsen et al., 2022; Sun and Sosnoff, 2018).

Wearable Inertial Measurement Units (IMUs) and machine-learning techniques provide a practical means of quantifying gait and functional movement (Chen et al., 2022; Subramaniam et al., 2022). However, much of the existing literature focuses on fall detection or one-off fall-risk classification rather than repeated,

participant-specific mobility assessment (Bet et al., 2019; Usmani et al., 2021). This project investigates a wearable multi-node IMU framework that combines activity recognition with participant-specific gait analysis and measurement of selected TUG-related movement phases, with the longer-term aim of supporting objective rehabilitation monitoring.

The Timed Up and Go test is particularly suitable for instrumented analysis because it combines several components of functional mobility in one task. A standard TUG requires an individual to rise from a seated position, walk approximately three metres, turn, return to the chair and sit down. It therefore incorporates postural transitions, straight-line walking, and turning performance (Podsiadlo and Richardson, 1991).

Conventionally, these movements are represented by a single completion time. Similar total TUG durations, however, can arise from different movement limitations. One individual may have difficulty rising from the chair, while another may rise easily but walk or turn more slowly. This distinction matters because gait speed, movement transitions, and other spatiotemporal characteristics have been associated with functional impairment and fall risk in older adults.

Instrumented TUG approaches therefore attempt to divide the test into measurable phases and extract information not represented by total completion time alone (Ortega-Bastidas et al., 2023). This allows TUG to be treated as a collection of interpretable mobility characteristics rather than only a stopwatch-based assessment.

IMU-based gait and TUG performance also make it possible to quantify mobility characteristics that may vary as physical function improves or deteriorates.

Gait speed is particularly relevant. The World Falls Guidelines recommend its use as part of fall-risk assessment and identify walking speed below 0.8 m/s as a useful reference for risk stratification in older adults (Montero-Odasso et al., 2022). More recent prospective evidence also supports an association between slower gait and subsequent falls, although gait speed alone has modest ability to distinguish between people who will and will not fall (Hicks et al., 2026).

Other characteristics provide complementary information. Increased stride-to-stride variability has been associated with future falls (Hausdorff, Rios and Edelberg, 2001), while cadence, step timing and sit-to-stand performance can provide additional information about mobility and lower-limb function.

These measures should not be interpreted individually as diagnostic indicators of fall risk. Their role in this project is to provide interpretable, participant-specific descriptions of mobility performance that could be compared across assessments and potentially used to monitor rehabilitation.

A continuous IMU recording contains acceleration and angular-velocity measurements, but these signals have limited meaning unless linked to the movement being performed. Gait characteristics, for example, should be calculated during periods of walking, while transition measures depend on recognising changes between movement states. Activity recognition therefore, forms an important intermediate stage in the proposed system.

Rather than attempting to infer fall risk directly from raw sensor signals, the framework first identifies sitting, standing and walking. These recognised states are then used to organise the sensor data into interpretable mobility events. Four machine-learning approaches are compared, with particular emphasis on their ability to generalise to unseen participants.

The resulting activity information provides the basis for estimating gait, step, and transition movement patterns and presenting these measurements within a real-time rehabilitation-monitoring prototype.

Despite growing interest in wearable activity recognition and sensor-based TUG assessment, considerable variation remains in sensor placement, signal-processing methods, movement segmentation and evaluation procedures (Sun and Sosnoff, 2018; Montesinos, Castaldo and Pecchia, 2018; Ortega-Bastidas et al., 2023). This methodological heterogeneity makes direct comparison between approaches difficult and limits confidence in how reliably findings generalise across participants.

Activity recognition and mobility assessment are often considered separately. However, movement states must first be identified reliably before activity-specific measures, such as cadence, step patterns and transitions, can be meaningfully derived.

This project therefore investigates an integrated three-node IMU framework in which participant-independent activity recognition supports subsequent gait and TUG-related analysis. It does not propose a clinical fall-risk classifier. Instead, it focuses on the technical reliability, interpretability, and real-time feasibility of a wearable system capable of producing participant-specific mobility measurements that could support repeated rehabilitation assessment.

Motivation

This research investigates how wearable IMU data can be transformed into interpretable measures of functional mobility. It addresses the need for more detailed gait and TUG assessment, while also considering the challenge of

extracting reliable activity and movement measures from continuous multi-sensor data.

By combining Human Activity Recognition with participant-specific mobility analysis, the study examines whether wearable sensing can provide objective information that complements conventional completion-time readings and supports repeated rehabilitation monitoring.

Aim

The aim of this project is to develop and evaluate a wearable multi-node IMU system for participant-independent recognition of key movement states and the extraction of interpretable gait and TUG-related measurements to support personalised rehabilitation monitoring.

Objectives

1. To develop a three-node wearable IMU acquisition system and use it to collect labelled acceleration and gyroscope data for select TUG-related movements.
2. To develop a reproducible pipeline for preprocessing, filtering, and segmenting multi-node IMU recordings for subsequent activity recognition and mobility analysis.
3. To develop and compare Random Forest, 1D-CNN, LSTM, and Conv-LSTM models for participant-independent recognition of sitting, standing, and walking.
4. To derive interpretable mobility parameters from recognised movement states and IMU signals, including step patterns, cadence and movement transitions relevant to gait and TUG assessment.

5. To evaluate the selected activity-recognition approach in terms of participant-level generalisation, sensor-channel contribution and inference latency, and integrate it into a real-time proof-of-concept mobility-monitoring system.

Research Questions

1. RQ1: How effectively can a three-node wearable IMU system distinguish sitting, standing and walking in unseen participants?
2. RQ2: How do Random Forest, 1D-CNN, LSTM and Conv-LSTM compare under participant-independent evaluation?
3. RQ3: Which sensor locations and channels contribute most strongly to activity recognition?
4. RQ4: Is the selected model computationally suitable for near-real-time operation?
5. RQ5: To what extent can activity recognition and IMU-derived movement measures provide a more detailed representation of selected gait and TUG-related mobility components than aggregate completion-time measures alone?

Scope and Limitations of the Research

This study focuses on the use of wearable inertial sensors for functional-mobility assessment. Sitting, standing and walking are treated as the main movement states used for gait and TUG-related analysis, while sit-to-walk is treated as a sequence of these movements and not as a separate activity class. The aim is to recognise these states reliably and use them to provide more detailed information about movement performance than overall task duration alone.

The study mainly evaluates the technical feasibility of the system, including generalisation to unseen participants, the contribution of individual sensor signals, the extraction of interpretable mobility information and operation within a real-time prototype. Rehabilitation monitoring provides the intended application context, but participants were not followed over time to evaluate changes in rehabilitation outcomes.

Although some of the mobility variables considered in this study have been associated with fall risk, falls were not used as a prediction target and no prospective fall outcomes were collected. The prototype should therefore be interpreted as a wearable-sensor mobility-assessment system and not as a diagnostic or clinically validated fall-risk prediction tool.

Dissertation Structure

The remainder of this dissertation is organised into five chapters.

Chapter 2 critically reviews literature on gait analysis, the Timed Up and Go test, wearable inertial sensing, Human Activity Recognition and machine-learning approaches for mobility assessment before identifying the research gap addressed by this study.

Chapter 3 presents the research methodology and system design, including the sensing architecture, data-collection protocol, preprocessing strategy, segmentation, feature engineering, model-selection approach and evaluation procedure.

Chapter 4 describes implementation of the wearable sensing system, data-processing pipeline, machine-learning models and real-time monitoring prototype.

Chapter 5 evaluates the system through participant-independent model performance, comparative model analysis, sensor-channel contribution, inference latency and interpretation of derived mobility variables.

Chapter 6 revisits the research questions and objectives, summarises the findings, discusses the limitations and identifies directions for future development and clinical validation.

CHAPTER 2

LITERATURE REVIEW

Introduction

This chapter reviews the literature relevant to the use of wearable sensors for assessing functional mobility. It explores conventional and sensor-based Timed Up and Go assessment, gait and postural transitions, wearable IMU sensing and sensor placement, Human Activity Recognition, machine-learning approaches, and participant-independent evaluation. The chapter also considers the strengths and limitations of existing approaches and uses these findings to identify the research gap addressed by the proposed multi-node mobility-analysis system.

Functional Mobility Assessment and the Timed Up and Go Test

Functional mobility describes a person's ability to perform movements required for everyday living, including transferring between sitting and standing, maintaining upright posture and walking safely. A decline in these abilities can affect independence and is relevant to broader assessment of falls and rehabilitation progress.

The Timed Up and Go test, introduced by Podsiadlo and Richardson (1991), provides a compact clinical assessment incorporating several of these abilities.

During the standard test, an individual rises from a chair, walks approximately three metres, turns through 180°, walks back to the chair, and sits down. The

overall completion time of the TUG test provides a simple and repeatable measure of functional mobility.

While the TUG's strength lies in its simplicity, requiring little equipment and combining multiple functional tasks, a lot of information is lost by recording only the total completion time. The total duration does not show how the time was distributed across the TUG's component activities.

Two individuals may complete the test in the same overall time while showing different limitations. One may rise slowly but walk relatively quickly, while another may perform the chair transfer efficiently but walk more slowly.

TUG performance, however, should also be distinguished from fall prediction. Barry et al. (2014) evaluated 25 studies and concluded that TUG had limited ability to predict future falls among community-dwelling older adults when used in isolation. At a threshold of 13.5 seconds, pooled sensitivity was 0.31 compared with specificity of 0.74.

The World Falls Guidelines recommend gait speed as the stronger gait-related measure for fall-risk stratification and report less consistent evidence for TUG-based prediction, but this does not make TUG an ineffective mobility assessment. TUG remains useful for assessing functional mobility; however, overall completion time should not be treated as equivalent to prospective fall risk, while wearable sensing can reveal movement components that total duration alone may hide.

Gait and Transition Characteristics as Digital

Mobility Indicators

Wearable sensor-aided mobility assessment can describe functional performance using several quantitative indicators rather than relying on a single task duration. These include gait speed, cadence, temporal variability, step patterns, and the timing or dynamics of postural transitions.

Gait Speed

Gait speed is one of the most established measures of functional mobility.

The World Falls Guidelines recommend gait-speed assessment for fall-risk stratification and use walking speed below 0.8 m/s as a practical reference threshold. However, gait speed remains one component of wider multifactorial assessment.

Hicks et al. (2026) analysed individual participant data from 28 prospective studies involving 7,608 participants and found that walking below 0.8 m/s was associated with greater risk of subsequently falling (RR 1.27) and a higher fall rate (IRR 1.54).

However, the threshold achieved only 35% sensitivity and overall diagnostic accuracy of 58%.

For an instrumented mobility system, gait speed is therefore better understood as an interpretable description of walking performance that may contribute to a wider mobility assessment.

Cadence and Gait Variability

Walking performance is not determined only by average speed. It is a function of multiple factors including Cadence.

Cadence describes stepping frequency, while temporal variability describes how consistently successive steps or strides are produced.

Prospective research has associated greater gait variability with subsequent falls. Hausdorff, Rios and Edelberg (2001), for example, followed 52 community-living older adults for one year and found greater stride-time variability among participants who later reported falling.

Verghese et al. (2009) similarly found associations between incident falls and several quantitative gait characteristics, including slower gait and greater variability in temporal gait measures.

Cadence has also received attention in wearable monitoring. In a study involving 432 community-dwelling older adults, every 10 steps/min increase in free-living cadence was associated with a 13.2% lower fall rate.

This was an association within that population, not a validated universal cadence threshold.

Unlike the World Falls Guidelines' 0.8 m/s gait-speed reference, there is no equivalent universally established cadence cut-off justifying classification of an individual as high or low risk based on cadence alone.

For this reason, cadence and step timing are treated as continuous descriptions of walking behaviour that could be compared between assessments.

TUG and Other Postural Transitions

Postural transitions provide another representation of functional mobility.

Rising from a chair requires lower-limb force production, coordination, balance control and movement of the trunk relative to the lower limbs.

Instrumented research suggests that transition-level measurements can provide information not represented by total task duration.

Doheny et al. (2011) used accelerometers positioned at the sternum and thigh during a five-times-sit-to-stand assessment and found significant differences between fallers and non-fallers in sit-to-stand time and several acceleration-derived variables.

Wearable-sensor reviews have similarly identified postural transitions as potentially informative aspects of mobility assessment.

Bezold et al. (2021) reported that one of the more accurate classification approaches among the studies reviewed used sit-to-walk and walk-to-sit information collected during daily life. This is relevant to the present study because sitting, standing and walking are included in the collected dataset. Recognising these movement states provides a basis for identifying transitions such as sit-to-stand, stand-to-walk, walk-to-stand and stand-to-sit.

These transitions are identified from changes in recognised movement states rather than independently labelled transition boundaries. This means they can be derived directly from the available data, unlike turning-related information, since no dedicated turning activity was collected.

Instrumented Timed Up and Go Assessment

Instrumented TUG, commonly referred to as iTUG, extends the conventional test by using sensing technologies to analyse movement during the assessment rather than measuring only total duration.

Ortega-Bastidas et al. (2023) reviewed 40 iTUG studies and found inertial sensing to be the most frequently used technology, appearing in 75% of included studies.

The literature also showed extensive use of automatic phase segmentation and extraction of variables from individual portions of the task.

Evidence suggests that this additional information can complement total TUG duration.

Weiss et al. (2011), using a lower-back accelerometer, derived measurements including sit-to-stand and stand-to-sit duration, number of steps, step duration and other gait-related variables. Several accelerometer-derived measurements differed between fallers and controls even where stopwatch-derived TUG time did not clearly distinguish the groups. This suggests that phase- and movement-specific measurements can provide information that may be obscured when TUG performance is represented only by total completion time.

The iTUG literature also reveals substantial methodological variation. Studies differ in sensor number and location, phase definitions, segmentation algorithms, participant populations and extracted variables. Ortega-Bastidas et al. (2023) found extensive variation even among systems intended to analyse the same clinical test.

Direct comparison is therefore difficult, and the output of an instrumented test depends heavily on how its sensing and processing system is designed.

Relevance to the Present Study

A clear distinction is needed between the literature discussed above and what is implemented in this dissertation. Although previous studies show that a complete iTUG assessment may include turning, this was not captured in the present dataset. No dedicated 180° turning trials were collected, and turning was not included as a target class for Human Activity Recognition. As a result, the system developed in this study should not be considered a complete automated TUG segmentation system.

The study focuses on components represented by the available activity data:

- sitting;
- standing;
- walking;

Recognition of these states also provides the basis for detecting changes between activities during real-time operation, including:

- sit-to-stand;
- stand-to-sit;
- stand-to-walk; and
- walk-to-stand.

The standard TUG provides the functional framework for interpreting these movements, while the present implementation focuses on selected TUG-related mobility components rather than complete automated TUG segmentation.

Turning remains an important area for future development, and this distinction prevents capabilities demonstrated by full iTUG systems from being incorrectly attributed to the current prototype.

Wearable IMUs for Functional Mobility Analysis

Wearable IMUs commonly combine tri-axial accelerometers and gyroscopes, enabling measurement of linear acceleration and angular velocity. Their small size and portability allow movement to be measured without a fixed laboratory motion-capture environment.

Reviews consistently identify wearable inertial sensing as a promising approach to quantitative gait and mobility assessment.

Subramaniam, Faisal and Deen (2022) describe IMUs as one of the most commonly used wearable technologies for measuring gait characteristics including cadence, step length and temporal gait phases. Patel, Pavic and Goodwin (2020) identified 149 gait and posture characteristics across 35 studies using wearable inertial sensors. However, having a sensor does not automatically make the resulting measurements meaningful.

Sun and Sosnoff (2018) found considerable variation in measured parameters, movement tasks, sensor sites and modelling techniques, preventing firm conclusions about the ability of sensing systems to predict future falls.

Montesinos, Castaldo and Pecchia (2018) similarly identified high heterogeneity in sensor location, task selection and extracted feature types.

These findings shift the question from whether IMUs can measure movement towards how sensing configuration and computational methodology affect the usefulness of the resulting data.

Sensor Placement and Multi-Node Measurement

Sensor placement is an important methodological decision because movement measured at different anatomical locations reflects different aspects of the activity being performed.

Lower-back sensors are common in gait and fall-related mobility research.

Patel, Pavic and Goodwin (2020) found that a single sensor on the lower posterior trunk could identify several gait and posture characteristics relevant to mobility assessment.

Bezold et al. (2021) similarly reported the lower back as the most common sensor location among studies included in their review.

This does not mean that the lower back is optimal for every movement.

Montesinos, Castaldo and Pecchia (2018) found that effective feature-location combinations depended on the task. Lower-back linear acceleration performed well during sit-to-stand and stand-to-sit assessment, while lower-limb sensing provided useful information during walking.

A multi-node system can therefore potentially capture complementary movement information.

Measurements near the trunk can characterise body-level posture and movement, while lower-limb sensors capture movement closer to joints involved in stepping and chair transitions.

The trade-off is additional engineering complexity.

Multiple devices must remain attached consistently, transmit data reliably and be aligned sufficiently well for combined analysis.

More sensors therefore do not automatically produce a better system.

The three-node configuration used in this study (lumbar, thigh and knee) is treated as an experimental design choice, not an established optimal configuration.

Later channel-contribution analysis investigates whether information from all three locations contributes meaningfully to activity recognition.

Human Activity Recognition as the Link Between Sensing and Mobility Analysis

Continuous IMU data are made up of multiple time-series signals recorded simultaneously. However, these measurements do not directly indicate whether the wearer is sitting, standing or walking.

Before activity-specific mobility measures can be calculated, the sensor stream therefore requires contextual interpretation.

Human Activity Recognition addresses this problem by estimating the activity associated with a segment of sensor data.

Wearable HAR commonly follows a sequence involving signal preprocessing, temporal segmentation, feature representation and classification. Sliding windows are frequently used to divide continuous recordings into manageable segments.

In this study, HAR acts as an intermediate step by recognising sitting, standing and walking from multi-node IMU signals before activity-specific mobility analysis is performed, rather than being used for fall-risk classification.

The target variable used to train the models is activity, not whether the participant subsequently falls.

Recognising walking provides the context in which step and cadence measurements can be considered.

Recognising sitting and standing provides the state information required to detect transitions between those postures.

HAR therefore connects raw sensing with interpretable movement events.

Turning is not included among the activity labels because no dedicated turning class was collected.

Machine-Learning Approaches for Wearable HAR

Wearable activity recognition can be approached using classical feature-based machine learning and deep-learning methods.

The primary difference is how useful representations of the sensor data are obtained.

Feature-Based Classical Machine Learning

Classical models generally operate on features extracted from each time window rather than the complete raw sequence.

Statistical descriptors such as mean, standard deviation, minimum, maximum, median, root-mean-square amplitude and temporal change can reduce a multichannel sequence to a structured feature vector.

Random Forests are useful in this context because they can represent nonlinear relationships, accommodate feature interactions and provide relatively efficient inference.

Because they rely on engineered features, their modelling pipeline can also be easier to inspect than some end-to-end deep-learning architectures.

The limitation is that the model can use only information represented by the selected feature set. Useful temporal structure may be lost when a time series is reduced to summary statistics.

Convolutional Approaches

One-dimensional Convolutional Neural Networks learn local temporal patterns directly from multichannel sensor sequences. Their convolutional filters can identify short patterns that appear repeatedly within the signals without requiring these patterns to be manually designed. For relatively short HAR windows, CNNs can also be computationally efficient because convolution operations can be processed in parallel.

Recurrent Approaches

Long Short-Term Memory networks are designed to learn dependencies across sequential data. This makes them relevant to human movement because observations at one point in time are connected to those that occur before and after them. However, the additional model complexity can increase computational cost and may require more training data to generalise effectively.

Hybrid Convolutional-Recurrent Models

Hybrid approaches combine convolutional feature extraction with recurrent sequence modelling. Reviews of deep learning in wearable HAR show that CNNs, recurrent networks and combined architectures have all been widely explored. However, no single architecture consistently performs best across different activities, sensor configurations and datasets. This makes direct comparison

important, as greater model complexity does not necessarily lead to better performance. Instead, the key question is whether a more complex architecture provides a clear advantage when all models are evaluated under the same participant-independent conditions.

Participant-Independent Evaluation

Evaluation methodology is especially important in wearable activity recognition because data collected from the same participant are often more closely related than data collected from different individuals. This difference becomes more significant when continuous sensor signals are divided into overlapping windows, since neighbouring windows may contain many of the same samples. If these windows are randomly split between training and testing sets, similar or partially overlapping data may appear in both, which can lead to an overly optimistic estimate of model performance.

Wearable HAR research therefore distinguishes between subject-dependent and subject-independent evaluation.

Scheurer et al. (2020) define subject-independent performance as recognition of users not represented in model training and identify Leave-One-Subject-Out (LOSO) validation as an appropriate way to estimate this capability.

More recent HAR research similarly warns that standard cross-validation applied to overlapping time-series windows can introduce data leakage and overestimate generalisation, recommending participant-level separation.

This point is central to the project. A rehabilitation-monitoring prototype should be able to process data from individuals whose movement patterns were not available during model development, so it is not enough to show that the model can recognise new windows from participants it has already seen. Leave-One-

Subject-Out cross-validation addresses this by training on $N-1$ participants and evaluating on the remaining participant, so that every participant is used once as the unseen test subject. This provides a more demanding estimate of generalisation than random window-level splitting and creates a consistent basis for comparing the Random Forest, 1D-CNN, LSTM and Conv-LSTM models.

Interpretability and Sensor Contribution

Interpretability is of great importance in a three-node system because the sensing architecture itself forms part of the investigation.

If one sensor location consistently contributes little useful information, future designs may be able to reduce hardware complexity by removing the node in question. Conversely, strong contributions from different anatomical locations would support multi-node sensing.

Channel importance should still be interpreted carefully. A signal that helps the model distinguish sitting from walking should not automatically be treated as a clinical biomarker of rehabilitation progress or fall risk; it only shows that the signal contains useful information for activity recognition.

This means that identifying which sensor signals contribute to activity recognition is different from determining which physiological variables predict future falls. The first question can be investigated using the HAR models developed in this study, while the second would require appropriate clinical outcomes and a different experimental design. In this project, channel-level importance is therefore used to understand model behaviour and evaluate the contribution of signals from the lumbar, thigh and knee sensor locations.

Computational Feasibility and Real-Time Monitoring

A model intended for live monitoring should be evaluated not only by classification performance, but also by how quickly it can generate predictions. This creates a trade-off between model complexity and practical deployment, since a more complex model may improve recognition but require greater computational cost, while a simpler model may offer lower latency. For a window-based system, near-real-time operation is feasible when inference time remains well below the interval between consecutive prediction windows.

The comparison therefore considers participant-independent performance alongside inference latency, interpretability and deployment complexity. The selected classifier is also integrated into a live proof-of-concept monitoring interface, so model selection is based on overall practical suitability rather than architectural complexity or a single performance metric.

Rehabilitation Monitoring and Personalised Mobility Assessment

Personalised rehabilitation monitoring is concerned not only with how an individual compares with population-level reference values, but also with how their mobility changes across repeated assessments. An individual may show meaningful improvement or decline over time even without crossing a predefined clinical threshold.

Wearable sensing is useful in this context because the same movement-related information can be collected repeatedly and associated with the same participant. For example, changes in estimated cadence or other validated mobility measures could be compared across successive assessment points to identify within-person trends over a rehabilitation period. This forms the basis

for the term “personalised” in the present study: the system produces participant-specific mobility information rather than relying only on population-level classifications.

The machine-learning model itself is not personalised through participant-specific retraining. Instead, it is developed using participant-independent evaluation so that it can be applied to individuals who were not represented during model development, while still producing mobility outputs associated with each participant.

However, participants in this study were not followed longitudinally through a clinical rehabilitation programme. The study therefore cannot determine whether the system can detect rehabilitation improvement or whether use of the prototype leads to better rehabilitation outcomes. Instead, it focuses on the technical foundations required for this type of monitoring, including wearable data acquisition, participant-independent activity recognition, extraction of interpretable mobility measures, and near-real-time presentation of results.

Longitudinal clinical validation remains an area for future work.

Critical Synthesis and Research Gap

The literature supports the use of quantitative mobility assessment but also highlights several limitations relevant to this research. Conventional TUG remains useful for assessing functional mobility, although total completion time reduces several movement components to a single value and should not be treated as a standalone predictor of future falls. Gait speed, cadence, temporal variability and postural transitions can provide additional information about mobility, but their associations with functional performance and fall-related outcomes do not make them independent diagnostic thresholds.

Wearable sensors can reveal movement characteristics that are not represented by overall TUG duration, although existing studies vary considerably in sensor placement, segmentation methods, phase definitions and extracted features. Multi-node IMU sensing may provide complementary movement information but also introduces additional acquisition and synchronisation complexity. Converting continuous sensor signals into meaningful mobility information therefore requires reliable activity recognition. Since no single machine-learning approach has demonstrated consistent superiority across wearable HAR applications, candidate models should be compared under equivalent participant-independent evaluation conditions rather than using randomly divided, highly correlated windows.

The contribution of this dissertation therefore lies not in the individual use of wearable IMUs, TUG analysis or HAR, but in their integration within a framework that:

1. acquires movement signals from multiple wearable locations;
2. recognises sitting, standing and walking in unseen participants;
3. compares classical and deep-learning models using the same participant-independent outer test folds;
4. examines the contribution of individual sensor channels;
5. derives interpretable walking measures and classifier-derived activity-state transitions from recognised movement states; and
6. evaluates whether the selected model can support near-real-time monitoring.

The scope remains narrower than that of a complete sensor-based TUG system because dedicated turning data were not collected and the study does not attempt full automatic segmentation of the standard TUG sequence. Instead, it focuses on selected gait and TUG-related mobility components supported by the

available activity data, while addressing participant-independent recognition, model comparison, sensor contribution, near-real-time feasibility, and whether recognised movement states can provide more detailed mobility information than aggregate timing alone.

CHAPTER 3

NEW IDEAS & APPROACH

Introduction

Chapter 2 identified several methodological limitations in wearable mobility and Human Activity Recognition (HAR) research, including differences in sensor placement, signal preprocessing and segmentation, participant-dependent evaluation, inconsistent definitions of instrumented Timed Up and Go (TUG) phases, and limited consideration of the computational requirements associated with live monitoring. This chapter presents the methodological framework developed to address these issues.

The study uses three body-worn inertial measurement unit (IMU) nodes positioned at the knee, thigh and lumbar regions. Following data-quality assessment, preprocessing and temporal segmentation, four classification approaches are compared: Random Forest (RF), one-dimensional Convolutional Neural Network (1D-CNN), Long Short-Term Memory network (LSTM) and Convolutional LSTM (Conv-LSTM). Generalisation to unseen participants is assessed using Leave-One-Subject-Out (LOSO) testing, while model performance is considered alongside sensor-channel contribution and computational inference latency.

The supervised HAR problem is restricted to three movement states:

$$Y = \{sit, stad, walk\}$$

Sit-to-walk-labelled recordings are retained separately because Sit-to-walk represents a composite function of th TUG assessment rather than an additional

mutually exclusive movement state. Since no dedicated turning activity was collected, the study does not attempt complete automatic decomposition of the conventional TUG sequence. The methodology also examines whether recognised movement states and IMU signals can provide interpretable information beyond categorical activity recognition. These outputs include estimated step count, estimated walking cadence, TUG-related recording duration and classifier-derived activity-state transitions. The selected classifier is subsequently incorporated into a near-real-time monitoring prototype to investigate whether sensing, preprocessing and inference can operate together within the intended prediction-update cycle.

The chapter therefore addresses participant-independent recognition, comparative model performance, sensor contribution, computational feasibility and the additional mobility information obtainable from the available signals. These forms of evidence are interpreted separately so that successful technical integration is not treated as equivalent to predictive or clinical validation.

Research Design

The study adopts a quantitative empirical research design. It combines controlled wearable-sensor data collection, comparative machine-learning experimentation and proof-of-concept system development. The methodology follows four connected stages:

1. wearable data acquisition;
2. dataset construction and preprocessing;
3. participant-independent HAR evaluation; and
4. mobility analysis and near-real-time deployment.

The first stage collects inertial data from the knee, thigh and lower lumbar regions. These independent sensor streams are then assessed for quality,

approximately aligned, filtered and segmented into a multichannel dataset.

Random Forest, 1D-CNN, LSTM and Conv-LSTM are subsequently evaluated to determine how well each approach generalises to individuals whose data were not used during model development.

Evaluation considers more than classification performance alone. Raw-channel permutation analysis is used to understand which sensor signals contribute to the model's predictions, while inference latency is measured to determine whether the candidate models can generate predictions within the intended update interval. Recognised movement states and IMU signals are also used to derive selected mobility information. The selected classifier is then integrated into a live prototype to assess whether the sensing, preprocessing and inference stages can operate together as a complete system.

These stages provide different forms of evidence. Classification experiments address predictive generalisation; model comparison addresses the relative suitability of alternative modelling approaches; permutation analysis provides interpretive evidence about sensor-channel reliance; latency benchmarking evaluates computational feasibility; and live deployment demonstrates technical integration. A functioning prototype therefore provides evidence that the components can operate together, but does not independently establish predictive accuracy or clinical validity.

WEARABLE IMU NODE – CIRCUIT DIAGRAM

Arduino UNO R4 WiFi + LSM6DS3 IMU

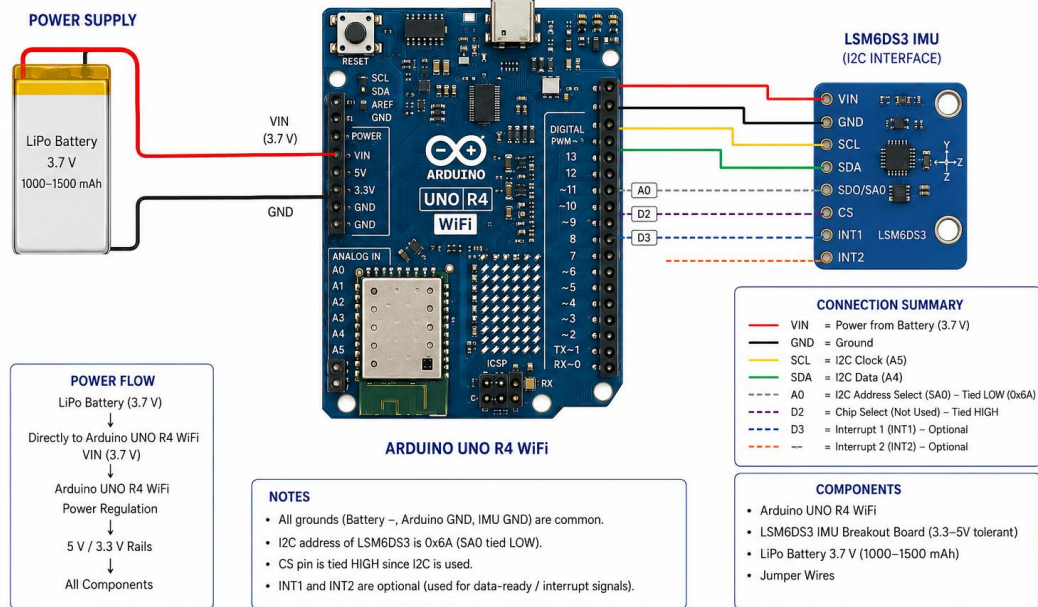


Figure 3.1: Overall methodological workflow of the proposed multi-node wearable mobility-analysis framework.

Relationship Between Research Questions and Experimental Design

Each research question is linked to a defined component of the experimental methodology.

Table 3.1: Relationship Between the Research Questions and Experimental Evidence

Research Question	Experimental Evidence
RQ1: How effectively can the three-node system recognise sitting,	Participant-independent LOSO evaluation

standing and walking in unseen participants?	
RQ2: How do Random Forest, 1D-CNN, LSTM and Conv-LSTM compare?	Common participant-level outer folds and evaluation metrics
RQ3: Which sensor locations and channels contribute most strongly to recognition?	Raw-channel permutation importance
RQ4: Is the selected model computationally suitable for near-real-time operation?	Single-window computational-latency benchmarking
RQ5: Can HAR and IMU-derived information provide more detail about selected gait and TUG-related mobility than aggregate timing alone?	Walking analysis, TUG-related recording duration, classifier-derived activity-state transitions and prototype outputs

This mapping ensures that each analytical procedure addresses an identified research question rather than becoming an isolated machine-learning experiment. Latency benchmarking, for example, is included because recognition performance alone cannot establish suitability for repeated near real-time inference. Sensor-channel analysis similarly investigates whether information obtained from the three-node arrangement contributes meaningfully to recognition rather than assuming that additional sensing locations justify their associated hardware and communication complexity.

A central methodological principle is that the participant, rather than the segmented window, is treated as the principal independent unit of

generalisation. Multiple windows from one participant constitute repeated observations and, because adjacent windows overlap, are not interpreted as independent experimental subjects. This distinction directly informs the validation strategy described in Section 3.9.

Sensing Strategy and Measurement Assumptions

The sensing configuration consists of three independent wearable nodes positioned at the knee, thigh and lumbar regions. Each node combines an Arduino Uno R4 WiFi-compatible microcontroller with an LSM6DS3 IMU providing tri-axial acceleration and angular-velocity measurements. Together, the three nodes provide a total of 18 raw IMU channels.

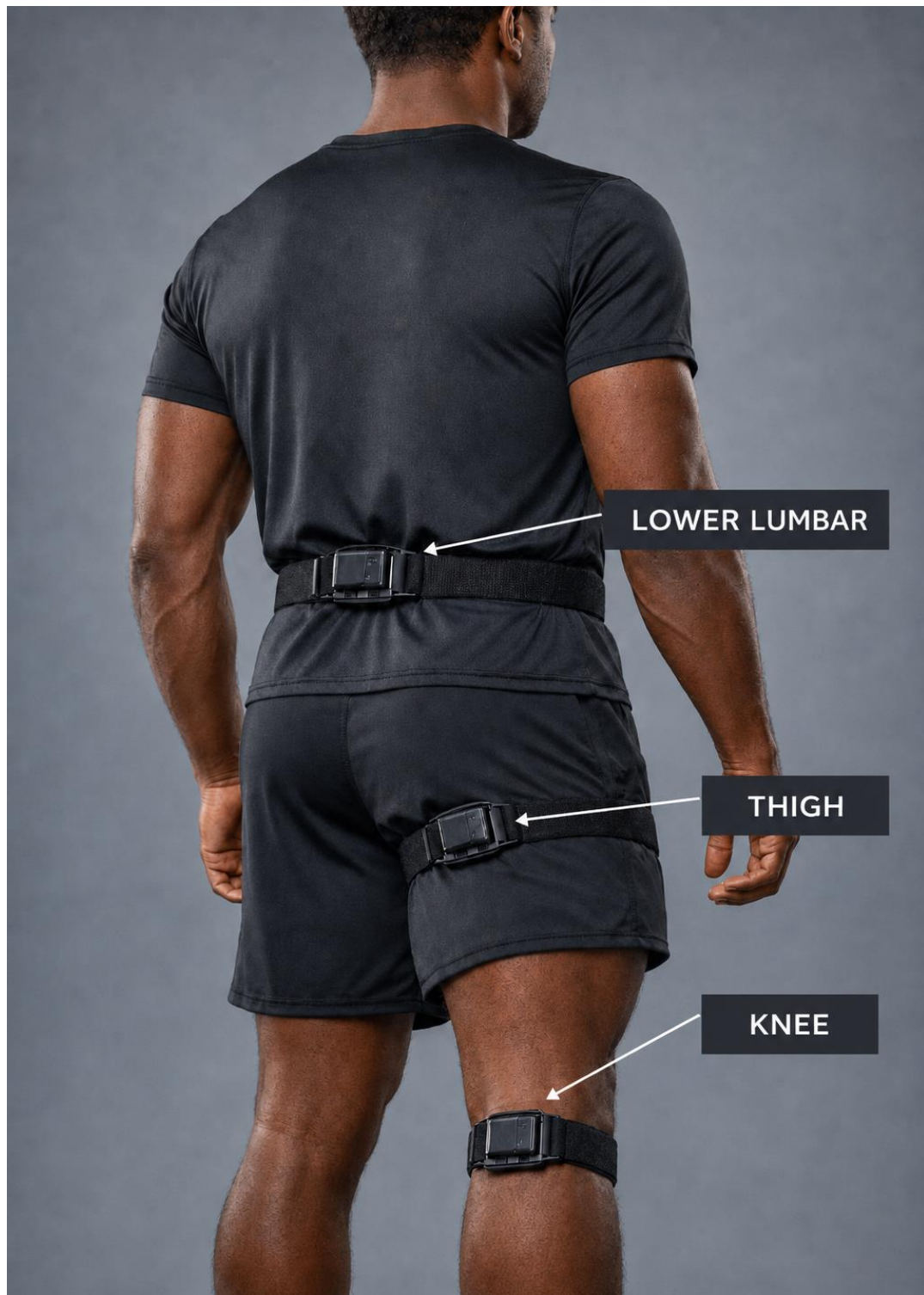


Figure 3.2: Placement of the Three Wearable IMU Sensors at the Lower Lumbar, Thigh and Knee Regions.

The three sensors were positioned at the lumbar, thigh and knee to capture movement from the trunk and lower limb during sitting, standing and walking.

The configuration is treated as an experimental sensing arrangement, rather than an established optimal placement. The later channel-contribution analysis therefore investigates how the different recorded signals influence activity recognition.

Sampling Configuration and Timing Uncertainty

The LSM6DS3 devices are internally configured at a higher output-data rate than the application-level acquisition schedule. At the application level, the firmware is configured to acquire observations at a nominal interval of:

$$Ts = 0.020s T_s = 0.020 \text{ s}$$

Corresponding to:

$$f_{configured} = 10.020 = 50Hz f_{configured} = \frac{1}{0.020} = 50 \text{ Hz}$$

The configured 50 Hz sampling rate is therefore used throughout the analysis for filtering, windowing and converting samples into time. However, quality-control checks on the recorded data showed a median interval of approximately 17 ms between successive Arduino timestamps:

$$\Delta t_{median} \approx 17ms$$

If these timestamps directly represent the physical acquisition interval, the corresponding effective sampling frequency would be approximately:

$$f_{observed} \approx 100017 \approx 58.8Hz f_{observed} \approx \frac{1000}{17} \approx 58.8 \text{ Hz}$$

The methodology therefore treats the configured 50 Hz sampling rate and the observed 17 ms timestamp interval as separate timing references. This

discrepancy has a more direct effect on outputs that depend on physical time, such as the physical duration represented by each analysis window, cadence, recording duration and transition-related timing, than on classification using fixed 100-sample sequences. Classification may nevertheless be affected indirectly because signal filtering is also performed using the configured 50 Hz sampling rate.

Until the discrepancy between the configured acquisition interval and the stored timestamp behaviour is resolved, 50 Hz is treated as the configured analytical time base, and time-derived outputs are interpreted subject to this uncertainty.

Data Collection and Raw Data Organisation

Participants are represented using pseudonymous identifiers rather than names.

Recordings are organised under four experimental labels:

1. Sit
2. Stand
3. Walk
4. Sit-to-walk

Sitting, standing and walking provide the activity recordings used for supervised HAR. Sit-to-Walk-labelled recordings remain separate because they contain multiple movement states and cannot be treated as a fourth mutually exclusive primitive activity and would introduce ambiguity into the classification target.

The recordings were collected under controlled conditions, so the HAR analysis focuses on differences in how participants performed sitting, standing and walking rather than free-living activity. Turning was not collected as a separate activity and is therefore not included as an HAR class.

Each sensor stream is preserved separately during acquisition. The stored data include server-side and Arduino-local timing information, sensor position, activity label, six IMU measurements and communication-quality information. Arduino timestamps represent the local clock of each microcontroller, while server timestamps represent data reception at the host computer. Because the three devices do not share the same clock, neither timestamp provides exact synchronisation across all three sensors.

The study involved eight volunteer participants aged between 26 and 50 years. Participation was limited to individuals who were able to perform the required movements safely. One participant had a walking impairment and was retained within the study population rather than excluded. Verbal consent was obtained from each participant before data collection. The recordings were conducted under controlled conditions, with each participant completing the sitting, standing, walking and TUG-related activities.

For the individual HAR activities, participants were instructed to sit comfortably on a chair with both feet on the ground for approximately two minutes, stand upright for approximately two minutes, and walk at a comfortable pace for approximately two minutes. For the Sit-to-Walk-labelled recording, participants initially remained seated for approximately five seconds before standing and beginning to walk. Because the collected TUG-related task did not include independently annotated phases or dedicated turning data, these recordings are treated as composite functional recordings rather than as a fully instrumented standard TUG protocol.

Three wearable sensor nodes were attached at the lumbar, thigh and knee regions. The IMU sensor on each node was secured to its Arduino board using double-sided tape to prevent movement of the sensor relative to the board during recording. The Arduino assembly was then attached to a Velcro belt,

which was fastened around the corresponding body region. Sensor orientation relative to the body was not standardised across participants. This allowed natural variation in device orientation to remain within the collected data rather than enforcing a fixed orientation during acquisition. The intention was to expose the recognition pipeline to some of the placement variability that may occur between individuals; however, the effect of orientation variability on generalisation was not evaluated independently and is therefore not claimed as a demonstrated source of improved performance.

Data Quality, Multi-Sensor Alignment and Analytical Time Base

Stream-Level Quality Control

Each sensor stream is assessed independently before the three streams are combined. Required fields are validated, sensor measurements are converted to numeric form, non-finite values are treated as unusable and observations lacking valid IMU measurements are removed.

For an activity recording containing N_K , N_L and N_T usable knee, lumbar and thigh observations respectively, stream balance is defined as:

$$B = \frac{\min(N_K, N_L, N_T)}{\max(N_K, N_L, N_T)}$$

A recording is retained only when:

$$B \geq 0.80 \text{ And } \min(N_K, N_L, N_T) \geq 100$$

The second condition ensures that even the shortest sensor stream contains enough data for at least one complete HAR window. Together, these criteria prevent incomplete recordings from being combined with much longer sensor

streams and ensure that sessions are excluded because of data quality rather than model performance.

Median positive differences between successive Arduino timestamps are also recorded for each node. The quality-control stage consequently retains information on participant, activity, node-specific sample counts, stream balance, timing diagnostics and inclusion status, creating an auditable record of how the analytical dataset was constructed. The final notebook implements the 0.80 balance threshold and 100-sample minimum directly in the session-quality decision.

Multi-Sensor Alignment

As mentioned earlier, the three microcontrollers do not share the same hardware clock, so their Arduino timestamps cannot be matched directly across devices. Server timestamps are also affected by network delay and batched transmission, which means they cannot be used for exact synchronisation.

The recordings are therefore aligned according to acquisition order. Each sensor stream is truncated to the length of the shortest stream:

$$N_{\text{aligned}} = \min(N_K, N_L, N_T)$$

Observations at the same acquisition position are then paired to form one 18-channel sample:

$$K_i \leftrightarrow L_i \leftrightarrow T_i$$

This approach assumes that:

1. the devices start recording at approximately the same time;
2. their sampling rates remain reasonably similar throughout each recording; and

3. no significant loss of samples occurs within any of the sensor streams.

Under these assumptions, samples with the same acquisition index are treated as approximately corresponding in time:

$$t_{K_i} \approx t_{L_i} \approx t_{T_i}$$

If one device misses samples or its clock drifts over time, the streams may become slightly misaligned. For this reason, acquisition-order alignment is suitable for the activity-recognition task in this study, but it should not be treated as precise sensor synchronisation.

Analytical Timeline

Following alignment, a uniform analytical timeline is reconstructed as:

$$t_i = t_0 + \frac{i}{f_s}$$

where:

$$f_s = 50\text{Hz}$$

is the configured analytical sampling frequency.

The first server timestamp is used as the starting point for the reconstructed timeline, while the remaining timestamps are generated using the assumed 50 Hz sampling rate. This means the timeline is used for analysis but does not confirm that every sample was physically recorded exactly 20 ms apart. Time-based outputs are therefore interpreted with the sampling uncertainty described in Section 3.4.1.

Signal Preprocessing and Temporal Segmentation

Signal Filtering

Each of the 18 IMU channels was filtered separately within its own recording using a fourth-order Butterworth low-pass filter. The filter used a cutoff frequency of:

$$f_c = 4\text{Hz}$$

This means that slower movement-related signal components were retained, while higher-frequency fluctuations and noise were reduced.

Because the analytical pipeline assumes a sampling frequency of 50 Hz, the corresponding Nyquist frequency is:

$$f_N = \frac{f_s}{2} = 25\text{Hz}$$

where f_s is the sampling frequency. The Nyquist frequency represents the highest frequency that can be represented correctly at the configured sampling rate.

Offline processing uses second-order sections with forward-and-backward filtering. Applying the filter independently within each recording prevents the end of one participant or activity session from being treated as temporally adjacent to the beginning of another.

The 4 Hz cutoff is treated as a fixed preprocessing choice rather than evidence that all physiologically relevant movement occurs below this frequency.

Offline and live filtering are not exactly the same. Offline preprocessing uses the complete recording, while live inference can only process the samples available within the current analysis window. This difference is therefore treated as a limitation of the live system rather than assuming that the offline filtering process is reproduced exactly in real time.

Separation of HAR and TUG Data

Following preprocessing, the dataset is separated according to analytical purpose. Sit, stand and walk are used as the supervised HAR classes, while Sit-To-Walk-labelled recordings are kept separate for visualisations. This avoids overlapping class definitions and keeps activity recognition separate from the broader analysis of functional mobility.

Temporal Segmentation

Continuous HAR recordings are segmented into windows containing:

$$N_w = 100$$

samples with a hop of:

$$N_{\text{hop}} = 50$$

samples.

Under the configured 50 Hz time base, these correspond nominally to two-second windows with 50% overlap and a one-second interval between consecutive windows.

Segmentation is performed independently within participant, activity and session boundaries, preventing windows from crossing recordings or individuals.

Adjacent windows share half of their observations; consequently, the number of

generated windows is not equivalent to the number of independent experimental units.

The overlap provides repeated temporal observations and increases prediction-update frequency, but also creates substantial correlation between neighbouring windows. For this reason, windows from one participant are never divided between training and testing in the principal evaluation procedure.

Input Representation and Candidate Models

All four model families are based on the same 18 physical IMU channels.

Additional variables including acceleration magnitude, jerk, approximate orientation and other exploratory motion descriptors are retained for exploratory or mobility analysis but are not introduced as additional common predictors in the four-model HAR comparison.

The comparison keeps the same sensor data, participants, segmentation process and test folds across all four models, while allowing each model to use the input format that best suits its design. Although the models do not receive the data in exactly the same form, they are all based on the same underlying IMU signals.

Random Forest receives a statistical representation of each window. Seven descriptors are calculated for each of the 18 sensor channels:

1. mean;
2. standard deviation;
3. minimum;
4. maximum;
5. median;
6. root-mean-square amplitude; and
7. mean absolute temporal change.

For channel xx:

$$RMS(x) = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

and:

$$MAC(x) = \frac{1}{N-1} \sum_{i=2}^N |x_i - x_{i-1}|$$

The resulting representation contains $18 \times 7 = 126$ predictors per window.

This representation was used because the Random Forest cannot use the temporal structure of a $100 \times 18100 \times 18$ sequence in the same way as the CNN or LSTM models. The 100 samples from each channel were therefore summarised using a fixed set of descriptive features, producing a more compact representation of each window instead of flattening all 1,800 sample values into a single high-dimensional vector.

Neural Representations

The 1D-CNN and LSTM receive the complete 100×18 temporal sequence. Conv-LSTM reorganises the same 100 observations into four 25-sample subsequences with an input representation of:

$$4 \times 1 \times 25 \times 18$$

The four approaches therefore represent different mechanisms for extracting predictive information from the same underlying sensor measurements.

Table 3.2:

Comparison of Model Input Representations and Methodological Roles

Model	Input Representation	Methodological Role
Random Forest	126 statistical predictors	Feature-based ensemble
1-D CNN	100×18	Local temporal-pattern learning
LSTM	100×18	Recurrent Sequence modelling
Conv-LSTM	$4 \times 1 \times 25 \times 18$	Hybrid convolutional and recurrent modelling

The comparison is used to determine which modelling approach performs best with the available data, rather than assuming that a more complex model will automatically produce better results.

Participant-Independent Model Development and Evaluation

Unit of Generalisation and LOSO Design

The main evaluation question is whether the classifier can recognise sitting, standing and walking in a participant whose data were not used during model development. For this reason, the participant is treated as the main unit of generalisation.

A random split of individual windows was not used because several windows come from the same participant, and neighbouring windows also overlap by 50%. Randomly dividing these windows could place very similar observations

from the same person in both the training and test sets, which could make model performance appear better than it really is for a completely new participant. This is why participant-level evaluation is important in HAR (Dehghani et al., 2019; Scheurer et al., 2020).

Leave-One-Subject-Out cross-validation was therefore used. In each evaluation round, one participant was kept completely separate for testing. From the remaining participants, one was used for validation and the others were used for training. This ensured that the training, validation and test groups contained different participants, so the test results reflected performance on an unseen individual.

Leakage Control

For neural models, numerical standardisation is fitted exclusively using training-participant observations:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

The same transformation is then applied to validation and test observations.

The outer test participant therefore contributes nothing to scaler estimation, class weighting, model fitting, early stopping or model selection. Validation decisions use a participant separate from both the training and test groups.

Participant identity is never supplied as a predictive input. It is retained only for fold construction and participant-level reporting.

Evaluation Metrics

Performance is evaluated using accuracy, precision, recall, class-specific F1, weighted F1 and macro F1, together with confusion matrices and participant-level results. Weighted F1 is useful when the number of samples differs between classes, while macro F1 gives each class equal importance. Participant-level results are also reported because an overall score can hide differences in how well the models perform for individual participants. This is especially important because some participants do not retain every activity after quality control, so class-level results are interpreted alongside the activities that are actually available for each participant.

Random Forest Optimisation and Model Roles

The Random Forest analysis is separated into three stages. The first Random Forest is included in the initial comparison with the deep-learning models using the same training, validation and test structure. After this comparison, a separate nested evaluation is used to determine whether hyperparameter optimisation improves the Random Forest performance. In each outer fold, one participant remains completely separate for testing, while the remaining participants are used for model development.

Within the development group, participant-separated GroupKFold validation is used during the randomised hyperparameter search. The search considers parameters such as the number of trees, maximum depth, split and leaf constraints, the number of features considered at each split and class weighting. To make the comparison fair, both the untuned and tuned Random Forest models are trained using the same development participants and then evaluated on the same unseen test participant.

After the nested evaluation is completed, a final Random Forest configuration is selected and trained using all available HAR participants for use in the live prototype. This final model is treated as a deployment model rather than another independent evaluation result.

Sensor Contribution and Computational Feasibility

Raw-Channel Permutation Analysis

Raw-channel permutation importance is used to understand how much each sensor signal contributes to a model's predictions. For each test fold, the model's weighted F1 score is recorded. Each sensor channel is then shuffled between test windows without changing the 100-sample sequence within each window. The model is tested again, and the drop in weighted F1 shows how much performance depends on that channel. This process is repeated three times for each channel, and the average reduction in performance is recorded.

For the Random Forest, the shuffled windows are converted into the same 126 statistical features used during classification, while the neural models use their usual scaling and input transformations. A large drop in performance means the model depends more on that channel, while a small drop does not necessarily mean the channel is unimportant because other correlated signals may provide similar information. The analysis is carried out on one representative fold, and the Random Forest and neural models are not fitted using exactly the same participant groups during this stage. As a result, the importance of each signal is only interpreted within each model and is not directly comparable across models.

Computational-Latency Evaluation

Computational feasibility is evaluated by measuring how long each model takes to process one complete HAR window. For the Random Forest, this includes statistical feature extraction and prediction, while the neural models include scaling and network inference, with the additional input transformation required for Conv-LSTM. After a short warm-up period, up to 200 windows are tested and the mean, median, standard deviation and 95th-percentile inference times are recorded.

The model is deemed suitable for near-real-time use if its inference time remains below the one-second interval between consecutive HAR windows. This benchmark only measures the time required for model processing, so additional delays from data transmission, window collection, server processing and displaying the results are not included.

Mobility Analysis and Evidence Boundaries

The study also examines whether recognised movement states and IMU data can provide useful mobility information beyond the activity class itself. Because these outputs are supported by different types of evidence, they are interpreted separately based on what the available data can actually support.

Table 3.3: Classification of Mobility Outputs Produced by the Prototype

Output Type	Information
Direct supervised movement states	Sit, stand and walk

IMU-derived mobility estimates	Estimated step count, estimated cadence and TUG-related recording duration
Classifier-derived application events	Sit-to-stand, stan-to-sit, stand-to-walk and walk-to-stand state changes

The available data do not establish validated physical step count, gait speed, walking distance, dedicated turning events, independently annotated biomechanical transition duration, prospective fall outcome or longitudinal rehabilitation improvement.

Activity-State Transitions

Separately collected sitting and standing recordings do not represent a continuously observed physical sit-to-stand transition. Physical transition duration is therefore not estimated by subtracting the end timestamp of one recording from the beginning of another.

During live monitoring, changes are instead identified when stable HAR predictions move between the supported states.

Estimated Step Count

Walking analysis uses the magnitude of the knee acceleration signal:

$$a_{\text{mag}}(t) = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2}$$

The signal is median-centred, and gait events are identified from peaks separated by approximately 0.30 s, with peak prominence adjusted according to the variation in each recording. Each detected peak is treated as an estimated

gait event, and the total is reported as the estimated step count. Because no manually counted or independently measured reference step count was collected, the resulting values are treated as estimates and not as validated step-detection measurements.

Estimated Cadence

Estimated cadence is calculated from the estimated step count and walking duration:

$$\text{Cadence} = \frac{N_{\text{estimated steps}}}{T_{\text{walking}}} \times 60$$

and is reported in estimated steps per minute. This allows walking recordings of different durations to be compared, although the result still depends on the accuracy of the gait-event detector and the assumed time base. No clinical cadence threshold is applied. Walking distance and gait speed are not calculated because no independent distance measurement or participant-specific stride length was collected.

Model Selection, Deployment and Reproducibility

The deployment model is selected based on generalisation, class-level results, inference latency, and interpretability. Model complexity is only considered useful where it provides a clear improvement in performance or practical use.

To reduce information leakage and improve reproducibility, participant identity is used only for grouping and reporting, while fixed settings and random seeds are used where possible to keep the experiments consistent.

The selected model is then integrated into a Python-based near-real-time monitoring application. This demonstrates that data acquisition, preprocessing,

inference and presentation can operate together as one system, but it does not provide an independent evaluation of live predictive accuracy. Detailed implementation of the firmware, server processing, model loading and user interface is presented in Chapter 4.

Professional, Social, Ethical and Legal Considerations

Clinical Interpretation and Participant Safety

The main professional risk is interpreting the system outputs more strongly than the data support. Recognising walking does not mean that a participant has a low fall risk, and estimated cadence, step count and activity-state transitions should not be treated as clinical diagnoses.

Errors in the HAR model can also affect the mobility information derived from its predictions. For example, if walking is classified incorrectly, gait-event analysis may be applied to the wrong section of data. The resulting mobility outputs should therefore be interpreted alongside the recognition performance of the model.

The prototype is intended to support mobility assessment, not replace clinical judgement. The wearable nodes should also be attached securely so that they do not restrict movement or create a trip hazard, especially in future studies involving people with reduced mobility or balance.

Privacy and Data Protection

Movement recordings constitute participant research data. Pseudonymous identifiers are therefore used in the dataset, while directly identifying information should remain separate.

Data access, retention and processing should follow the approved institutional research procedure and applicable UK data-protection requirements.

Representativeness and Fairness

Participant-independent evaluation reduces subject leakage but does not establish population-wide clinical validity. Performance may vary according to age, body morphology, mobility impairment, pathological gait, mobility aids, sensor attachment and individual movement strategy. Therefore, future clinical evaluation would require broader recruitment and appropriate subgroup analysis.

Software Security and Intended Use

Network credentials and development passwords should not be included in publicly distributed firmware or source code. A production health-monitoring system would also require stronger authentication, encrypted communication, access control and secure credential management than the local research prototype.

The system developed in this study is not:

- a regulated medical device;
- a diagnostic system;
- a validated prospective fall-risk predictor; or
- evidence that rehabilitation outcomes improve through its use.

Any future clinical deployment would therefore require substantially wider technical, clinical and regulatory validation.

Threats to Methodological Validity

The principal limitations are grouped according to the type of inference they constrain.

Construct Validity

Construct validity describes whether or not the outputs produced by the system accurately represent what they are intended to describe. In this study, classifier-derived transitions are not direct measurements of biomechanical transition duration. Detected gait events have not been validated against a reference step count, and estimated cadence depends on both gait-event detection and the assumed time base. Sit-to-stand-related recording duration also does not represent a fully segmented clinical TUG test. Walking distance, gait speed and prospective fall risk are not directly measured.

For this reason, the outputs are described using qualified terms such as estimated step count, estimated cadence, classifier-derived activity-state transition and Sit-to-Walk-related recording duration.

Internal Validity

The three wearable nodes are not hardware synchronised, and acquisition-order alignment consequently provides approximate rather than exact temporal correspondence. Hidden sample loss, device-clock drift and uncertainty concerning the physical sampling interval may introduce further timing error.

Offline and live filtering also differ because complete-session zero-phase filtering cannot be reproduced exactly when future observations are unavailable during live inference.

These factors do not necessarily prevent broad activity recognition, but they constrain conclusions requiring precise physical timing or inter-sensor temporal relationships.

Statistical Conclusion Validity

Although segmentation produces a large number of windows, the number of independent participants remains small. Because neighbouring windows overlap, they are treated as repeated observations from the same participant and not as independent subjects.

Some held-out participants also do not retain every HAR class after quality control, so class-level results are interpreted alongside the activities available for each participant. Permutation analysis is based on one representative fold, and the Random Forest and neural models use slightly different fitting procedures, so sensor-contribution results are interpreted mainly within each model.

External Validity

Activities were recorded under controlled experimental conditions rather than unrestricted free-living behaviour. The HAR vocabulary is restricted to sitting, standing and walking, and no dedicated turning activity was collected.

Strong LOSO performance therefore supports participant-independent feasibility within the available dataset, but does not establish equivalent performance in broader clinical populations, free-living environments or substantially different sensor-placement conditions.

Deployment Validity

The final deployment Random Forest is trained using all available HAR participants after participant-independent evaluation. It is therefore an operational model rather than an additional independent test model.

The live application has also not undergone a separately annotated real-time accuracy experiment. Successful acquisition, inference and presentation demonstrate technical integration, but do not independently establish deployed predictive accuracy.

Chapter Summary

This chapter presented the methodology used to investigate participant-independent activity recognition and selected gait and TUG-related mobility analysis using three wearable IMU nodes positioned at the knee, thigh and lumbar regions. The independently recorded sensor streams were quality-assessed, approximately aligned and processed into an 18-channel representation for analysis.

Sitting, standing and walking formed the supervised HAR classes, while Sit-to-walk-labelled recordings were analysed separately as composite functional recordings. Random Forest, 1D-CNN, LSTM and Conv-LSTM were evaluated using participant-independent LOSO testing, with model development isolated from the held-out participant. Random Forest optimisation was subsequently assessed using a nested participant-wise procedure.

The methodology also incorporated sensor-contribution analysis, computational-latency assessment and selected mobility measures. These analyses were deliberately bounded by the available evidence, distinguishing model-derived outputs from validated clinical gait or TUG measurements. The resulting

framework therefore provides a controlled progression from wearable sensing to participant-independent evaluation and proof-of-concept deployment

CHAPTER 4

IMPLEMENTATION

Introduction

Chapter 3 described the methodology used to evaluate activity recognition and selected gait and TUG-related mobility outputs. This chapter explains how that methodology was implemented in hardware and software.

The implementation has three main parts: the three wearable sensor nodes placed at the knee, thigh and lumbar regions, the offline processing pipeline used to prepare the data and develop the models, and the near-real-time monitoring application used to process incoming sensor data with the selected model.

The chapter covers the software environment, sensor firmware, wireless data collection, dataset preparation, model implementation and deployment process. It also describes the live prediction process, activity-state transitions, step estimation and dashboard. The experimental results and their interpretation are presented in Chapter 5.

Implementation Environment and Software

Organisation

The analytical pipeline was implemented in Python using a Jupyter notebook. The main experimental settings are defined centrally, including the configured 50 Hz analytical sampling rate, knee, lumbar and thigh sensor locations, 100-sample HAR window, 50% overlap, filtering parameters, neural-network training

settings and Random Forest search space. A random seed of 42 was fixed across Python, NumPy and TensorFlow to improve repeatability.

The implementation uses pandas and NumPy for data handling and numerical operations, SciPy for signal filtering and peak detection, and scikit-learn for Random Forest modelling, scaling, cross-validation, hyperparameter optimisation and evaluation. TensorFlow/Keras is used for the neural-network models, while Matplotlib and Seaborn support visualisation. Model artefacts are serialised using joblib.

The near-real-time application was implemented separately from the experimental notebook so that the trained model could operate without running the complete development environment. aiohttp handles WebSocket communication with the wearable nodes and HTTP communication with the browser dashboard, while NumPy and SciPy support live preprocessing and mobility calculations. The wearable nodes were programmed in Arduino C++ using Wire, WiFiS3, WebSocketsClient, the LSM6DS3 library and ArduinoJson. The microcontrollers therefore perform sensing and transmission, while the more computationally demanding processing is carried out on the host computer.

Wearable Node and Firmware Implementation

Common Firmware and IMU Configuration

The three wearable nodes use the same firmware structure. Each device is identified as the knee, lumbar or thigh node through the `SENSOR_PLACEMENT` configuration, while sampling, buffering, IMU configuration and network communication remain common across all nodes.

As shown in Figure 4.1, the IMU was connected to the Arduino through the SDA and SCL lines, with common power and ground connections, while the same node design was replicated across the three body locations.

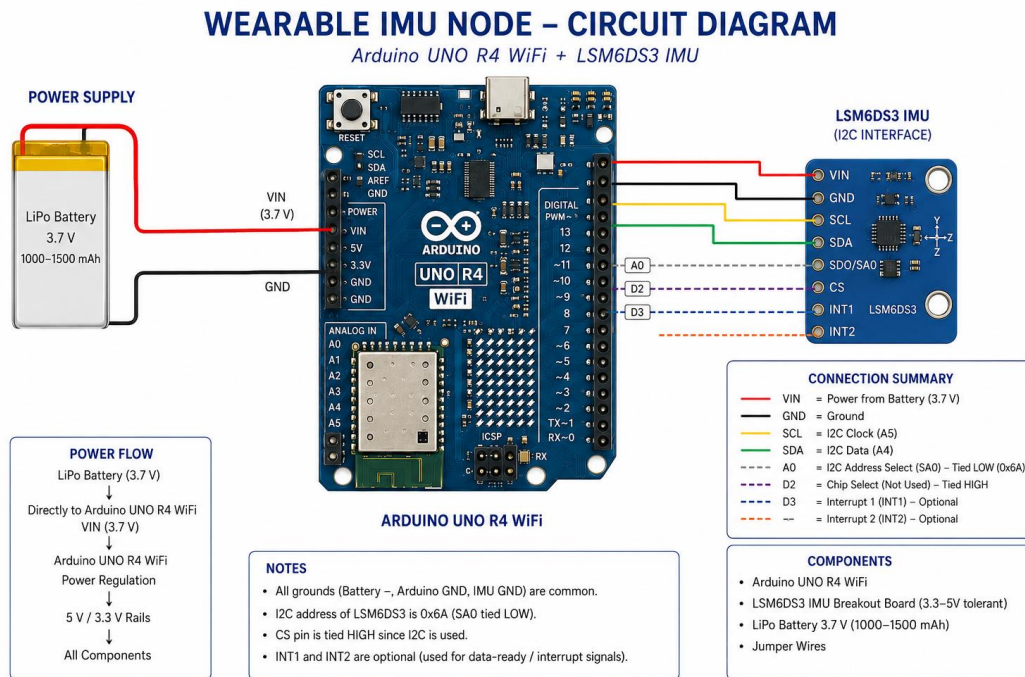


Figure 4.1: Circuit diagram of a wearable IMU node based on the Arduino Uno R4 WiFi and LSM6DS3 sensor. The same wiring arrangement was used for the lumbar, thigh and knee nodes, with node identity determined in firmware

Each node uses an LSM6DS3 IMU connected through a 400 kHz I²C interface. The accelerometer is configured to ± 16 g and the gyroscope to $\pm 2000^\circ/\text{s}$, with both configured internally at 104 Hz. If IMU initialisation fails, execution stops so that invalid measurements are not transmitted. Following successful initialisation, the node connects to Wi-Fi and establishes a WebSocket connection with the Python application through port 5001.

Each acquired observation contains an Arduino-local timestamp, tri-axial acceleration (a_x , a_y , a_z), tri-axial angular velocity (g_x , g_y , g_z) and RSSI. These values are temporarily stored in a firmware `SensorReading` structure before transmission.

Sampling, Buffering and Recording Control

Application-level acquisition is configured at a nominal interval of 20,000 μ s, corresponding to 50 Hz. The firmware compares `micros()` with the scheduled acquisition time and reads the six IMU channels when the required interval has elapsed. Under normal operation, the acquisition schedule advances by a further 20,000 μ s. If execution falls substantially behind schedule, the timer is reset to the current time rather than generating several rapid catch-up observations.

Ten observations are buffered before transmission, corresponding nominally to 0.2 seconds of data. The buffer is serialised as a JSON message containing the node identity and its timestamped observations before transmission through WebSocket. This reduces the number of network messages while keeping sensor acquisition separate from transmission.

Recording is controlled through `START` and `STOP` commands issued by the Python application. `START` activates the recording session, clears the existing buffer and resets the acquisition schedule, while `STOP` ends recording. Sensor observations are therefore collected only while the WebSocket connection is active and a session has been started, preventing unlabelled data from being collected between trials.

Data-Collection Server and Raw Data Storage

The Python application operates in three states: **Idle**, **Data Collection** and **Inference**. It waits for the three wearable nodes to connect before presenting the normal operating interface.

During data collection, the operator is first presented with an automatically generated participant identifier, which can either be accepted or replaced with a custom identifier. The operator is then prompted to specify the activity being recorded. Although sitting, standing, walking and TUG are presented as the intended activity options, a custom activity label can also be entered. The server creates the corresponding participant/activity directory and broadcasts a START command to the connected wearable nodes. Recording continues until the operator presses Enter, after which STOP is broadcast, the associated files are flushed and closed, and the application returns to idle mode.

```
data/
|__ user_1
|   |__ sit
|       _ session-knee.csv
|       _ session-lumbar.csv
|       _ session-thigh.csv
|   |__ sit-to-walk
|       _ session-knee.csv
|       _ session-lumbar.csv
|       _ session-thigh.csv
|   |__ stand
|       _ session-knee.csv
|       _ session-lumbar.csv
|       _ session-thigh.csv
|   |__ walk
|       _ session-knee.csv
|       _ session-lumbar.csv
|       _ session-thigh.csv
|__ user_2
|   |__ sit
|       _ session-knee.csv
|       _ session-lumbar.csv
|       _ session-thigh.csv
|   |__ sit-to-walk
|       _ session-knee.csv
|   ...
|   |__ walk
|       _ session-knee.csv
|       _ session-lumbar.csv
|       _ session-thigh.csv
```

Figure 4.2: Image showing the final folder structure for acquired data

A separate CSV file is maintained for the knee, lumbar and thigh nodes. Each row contains the server reception timestamp, Arduino-local timestamp, sensor position,

activity label, six IMU measurements and RSSI. Recordings are organised by participant and activity so that the analysis notebook can discover them automatically rather than relying on a manually defined file list. Data is flushed to persistent storage during acquisition to reduce the amount of information held only in memory.

Dataset Construction and Preprocessing

Recording Discovery, Cleaning and Quality Control

The notebook searches the participant and activity directories and identifies the expected knee, lumbar and thigh recordings. Sessions missing one or more required sensor files are not partially combined.

Each sensor stream is loaded independently and checked for the required timestamps and six IMU channels. Numerical fields are converted to appropriate types, non-finite values are treated as missing, and observations without complete IMU measurements are removed. A sequential sample index preserves acquisition order. The median positive interval between consecutive Arduino timestamps is also calculated for each node as a timing diagnostic.

The stream-quality criteria defined in Section 3.6.1 are implemented directly in the dataset-construction pipeline. Before alignment, the notebook calculates the usable sample count for each node and excludes recordings that fail either the stream-balance requirement or the minimum 100-sample requirement. A separate quality record stores the participant, activity, node-specific sample counts, aligned sample count, stream balance, median Arduino timestamp interval for each node and final inclusion status. This preserves an auditable record of the decisions used to construct the analytical dataset.

Multi-Node Alignment and Analytical Timeline

The alignment procedure defined in Section 3.6.2 was implemented by trimming each accepted sensor stream to the length of the shortest stream and concatenating corresponding knee, lumbar and thigh observations by acquisition order. This produces the 18-channel analytical representation used by the modelling pipeline.

The implementation then adds participant, activity and session identifiers and reconstructs the configured analytical timeline described in Section 3.6.3. Accepted recordings are combined into Dataset_A, while alignment-quality information is retained separately in session_sync_quality.csv.

Filtering and Analytical Dataset Separation

Each of the 18 IMU channels is filtered independently within its session_id using a fourth-order Butterworth low-pass filter with a 4 Hz cutoff and the configured 50 Hz analytical sampling rate. The filter is implemented using second-order sections and sosfiltfilt, preventing filtering across unrelated recording boundaries. The unfiltered data are retained separately, while the filtered data form the main analytical dataset.

Sitting, standing and walking recordings are used for HAR, while Sit-to-Walk-labelled recordings are retained separately for descriptive mobility analysis. Additional quantities, including acceleration magnitude, approximate orientation, jerk and angular-velocity descriptors, are generated for exploratory analysis but are not included in the final HAR predictor set, which uses the original 18 aligned IMU channels.

Before modelling, the pipeline checks channel availability, participant and session consistency, chronological ordering, finite numerical values, HAR class

labels, and the exclusion of target-derived and quality-control fields from the predictors.

HAR Windowing and Participant-Level Evaluation

HAR recordings are segmented separately within participant, session and activity boundaries into windows containing 100 observations across the 18 IMU channels. Consecutive windows use a 50-sample hop. The segmentation process produces temporal sensor windows, activity labels, participant identifiers and session identifiers for subsequent evaluation.

Activity labels are encoded separately from the sensor input. Participant-independent folds are created using `LeaveOneGroupOut`, with participant identity used as the grouping variable. In each outer fold, one complete participant is reserved for testing. A different participant from the remaining group is used for validation, while the remaining participants form the training set. The same participant-level folds are reused across the four candidate models.

For the neural architectures, `StandardScaler` is fitted using training-participant data only. Balanced class weights are also calculated from training labels only. Shared training utilities provide early stopping based on validation loss, restoration of the best network weights and learning-rate reduction. A common evaluation function calculates accuracy, weighted precision, weighted recall, weighted F1 and macro F1 so that all candidate models use consistent metric definitions.

Candidate Model Implementation

The four candidate models use the same underlying 18 IMU channels but represent each HAR window differently

Table 4.1: Model Architectures and Implementation Details

Model	Implementation
Random Forest	Each 100×18 window is transformed into 126 predictors using mean, standard deviation, minimum, maximum, median, RMS and mean absolute sample-to-sample change for each channel. Feature scaling is not applied.
1D-CNN	Receives the scaled 100×18 sequence and applies Conv1D(64, kernel 5), batch normalisation, max pooling, dropout, Conv1D(128, kernel 3), batch normalisation, global-average pooling, Dense(64), dropout and a three-class softmax output.
LSTM	Receives the scaled 100×18 sequence through LSTM(64), dropout, LSTM(32), Dense(32), dropout and a three-class softmax output.
Conv-LSTM	Reshapes each window into $4 \times 1 \times 25 \times 18$ before applying ConvLSTM2D(32), batch normalisation, dropout, flattening, Dense(64), dropout and a three-class softmax output.

The neural models use Adam optimisation with sparse categorical cross-entropy and follow the same preprocessing and validation process within each fold. The Random Forest converts the training, validation and test windows into the same 126 statistical features before the model is fitted.

Evaluation, Sensor Contribution and Model Optimisation

A shared LOSO evaluation function runs the predefined participant folds for each model and records the training, validation and held-out test results. For each fold, the pipeline stores the classification metrics, training time and training-to-test generalisation gap. Test labels and predictions are also combined across folds to produce overall model summaries, class-specific reports and confusion matrices. These results are presented and discussed in Chapter 5.

Raw-channel permutation importance is calculated using one representative LOSO fold. Each of the 18 sensor channels is shuffled between held-out windows while keeping the 100-sample sequence within that channel unchanged. Weighted F1 is recalculated after each shuffle, and the process is repeated three times per channel. The Random Forest uses the same 126-feature extraction process on the modified windows, while the neural models use their usual scaling and reshaping steps. The drop in weighted F1 is used to show how much the model depends on each channel.

After the initial comparison, the Random Forest is further evaluated using nested participant-level optimisation. In each outer LOSO fold, the training and validation participants are combined into a development set, while the test participant remains completely separate. RandomizedSearchCV uses participant-grouped GroupKFold validation and weighted F1 to test 12 configurations covering the number of trees, maximum depth, minimum split and leaf sizes, features considered at each split and class weighting. The baseline and optimised Random Forest models are then tested on the same held-out participant. A final consensus configuration is trained on all available HAR participants for deployment and is not used as an additional LOSO result.

Computational performance is measured using up to 200 HAR windows after 20 warm-up runs. `time.perf_counter_ns()` is used to record the mean, median, standard deviation and 95th-percentile latency. Random Forest timing includes feature extraction and prediction, while neural-model timing includes scaling and inference, with input reshaping also included for Conv-LSTM. For the neural models, this benchmark measures the computational cost of the implemented architecture and not the latency of a specific trained LOSO fold.

Mobility Analysis Implementation

SitTo-Walk-Related Recording Duration

Recording duration is calculated for each participant, session and activity combination using the reconstructed analytical timeline, including an estimate of the final sampling interval. SIT-to-walk-labelled recordings are extracted into a separate summary for descriptive analysis.

The implementation does not calculate individual Sit-to-Walk-phase durations and does not estimate physical sit-to-walk duration by combining separately collected sitting and standing recordings.

Estimated Step Count and Cadence

Walking analysis uses the magnitude of the knee acceleration signal to estimate gait events. The signal is median-centred before peaks are detected using `find_peaks`. Peaks must be at least 15 samples apart, which corresponds to approximately 0.30 seconds at the configured 50 Hz sampling rate. Peak prominence is adjusted using the variation within each recording.

Each detected peak is treated as an estimated step, and cadence is calculated from the estimated number of steps and the duration of the walking recording. The results are stored for each walking session and checked using peak-detection plots. Walking distance and gait speed are not calculated because participant-specific step or stride length was not measured.

Deployment and Near-Real-Time Inference

Deployment Artefact and Runtime Configuration

Following optimisation, the final Random Forest estimator and activity-label information are exported for use by the live application. The deployment process requires a consistent runtime configuration describing the activity classes, input-feature ordering, configured sampling rate, window size, hop size, filter parameters and sensor ordering. The server checks these settings against its expected configuration when the model is loaded.

The live pipeline therefore expects a configured 50 Hz analytical sampling rate, 100 observations per window, a 50-observation hop and a fourth-order Butterworth filter with a 4 Hz cutoff. Sensor channels are ordered as knee, lumbar and thigh, with acceleration followed by angular-velocity measurements for each node.

Live Inference Engine

The InferenceEngine maintains separate queues for incoming knee, lumbar and thigh observations. Each sensor message is checked for the six required numerical motion channels. Once observations are available from all three queues, one sample is removed from each and combined in the required 18-channel order.

The combined observations populate a rolling 100-sample buffer. The first prediction is generated when the buffer is full, followed by another prediction after every 50 newly aligned observations. Each completed window is filtered using the configured fourth-order Butterworth low-pass filter with a 4 Hz cutoff. The same seven statistical descriptors used by the offline Random Forest are then calculated to produce the 126-feature input vector.

The feature vector is passed to `predict()` and, where supported, `predict_proba()` to obtain the predicted activity, confidence and class probabilities. Reported live latency covers window preprocessing and model computation and does not include the time required to accumulate a complete input window.

Activity Stabilisation, Transitions and Live Step Estimation

Individual predictions are stabilised before being interpreted as activity changes. When a prediction differs from the current stable state, it becomes a candidate and must appear in two consecutive prediction windows before being accepted. If the prediction returns to the existing stable state before confirmation, the candidate is cleared.

The application tracks four supported state changes:

- sit to stand;
- stand to sit;
- stand to walk; and
- walk to stand.

For each accepted transition, the application records the previous state, new state, candidate start time, confirmation time and detection duration. The recorded duration represents the time required to confirm the classifier state change, not the physical duration of the movement.

Live step estimation is active only while the stable activity is walking. Knee acceleration magnitude is median-centred and processed using the same peak-spacing and prominence principles as the offline analysis. Because consecutive inference windows overlap by 50%, only peaks within the newly introduced half of later windows are added after the first prediction. This prevents peaks already counted in the previous window from being counted again. The resulting value remains an estimated walking step count at best.

Server and Browser Dashboard

A single aiohttp application manages communication with both the wearable nodes and the browser. WebSocket traffic is routed to the Arduino handler, while standard HTTP requests provide the dashboard. The `/api/state` endpoint returns the current application state as JSON, and `/health` provides a lightweight diagnostic endpoint. Incoming sensor messages are directed to CSV persistence during data collection or to the InferenceEngine during live inference.

The browser dashboard is implemented using HTML, CSS and JavaScript. It displays sensor connectivity, predicted activity, confidence, class probabilities, window-buffer status, estimated walking steps, elapsed experiment time, detected transitions, transition counts and durations, inference latency and relevant acquisition settings. The browser requests `/api/state` every 150 ms so that updated results can be displayed shortly after inference becomes available.

The application server and terminal interface operate within the same asynchronous event loop. During shutdown, open collection files are closed, STOP is transmitted where possible, active WebSocket connections are terminated and the aiohttp application is cleaned up.

Implementation Boundaries

The prototype only reports outputs that are supported by the collected data.

Turning is not classified because no dedicated turning recordings were collected, and separately recorded sitting and standing trials are not combined to estimate physical sit-to-stand duration. Walking distance and gait speed are also not calculated because participant-specific distance or stride information was not available.

Step count remains an estimate because it was not validated against an independent reference system. Transition duration also reflects how long the classifier takes to confirm a change in activity state, not the physical duration of the movement itself. Although the system is designed to support rehabilitation and fall-related mobility assessment, the model only predicts sitting, standing and walking, and does not directly predict fall risk.

Chapter Summary

This chapter described the implementation of the wearable mobility-analysis system from sensor acquisition to near-real-time deployment. Three LSM6DS3-based wearable nodes collect knee, thigh and lumbar IMU data and transmit them to a Python application, where recordings are quality-assessed, approximately aligned, filtered and segmented for HAR.

Random Forest, 1D-CNN, LSTM and Conv-LSTM were implemented within a common participant-level evaluation framework, followed by sensor-contribution analysis, Random Forest optimisation and computational-latency benchmarking. The selected Random Forest was then incorporated into a near-real-time application providing activity predictions, classifier-derived state transitions and estimated walking steps through a browser dashboard. These outputs remain

constrained by the evidence available from the collected data and are not treated as clinically validated measurements.

CHAPTER 5

EVALUATION

Introduction

This chapter evaluates the performance and practical suitability of the wearable mobility-analysis system developed in Chapters 3 and 4. The evaluation considers the constructed multi-sensor dataset, participant-independent recognition of sitting, standing and walking, Random Forest optimisation, sensor-channel contribution, exploratory Sit-to-Walk and walking measures, and computational suitability for near-real-time operation.

Particular attention is given to performance across unseen participants rather than aggregate accuracy alone. Activity recognition, mobility measurement and clinical interpretation are treated separately: the supervised models recognise sitting, standing and walking, while derived step counts, cadence, TUG-related duration and activity transitions remain experimental mobility outputs, and not validated clinical biomarkers.

Evaluation of the Dataset and Experimental Setup

Data were collected using the three wearable nodes positioned at the lumbar, thigh and knee regions. Photographs of the completed hardware and its placement on participants provide a physical reference for the sensing configuration evaluated in this chapter.

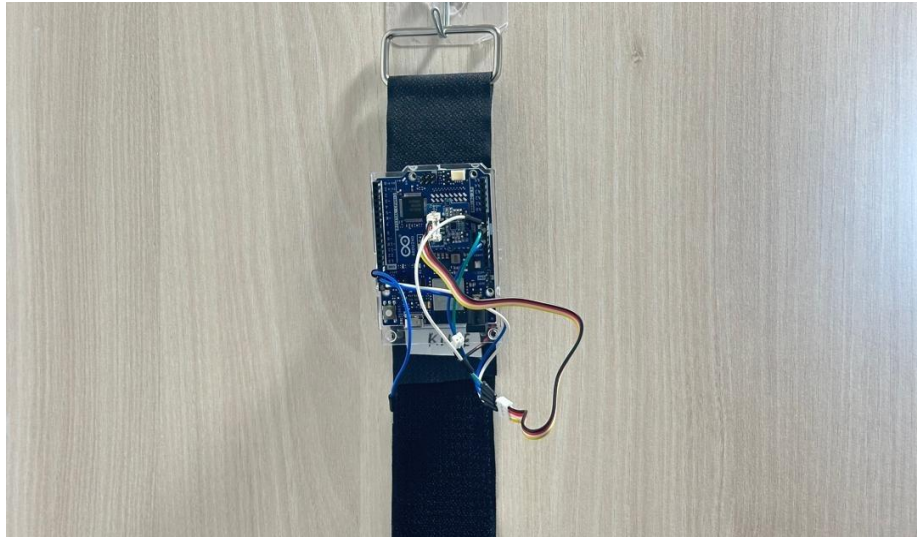


Figure 5.1: Example of a completed wearable IMU node used during data collection



Figure 5.2: Example placement of the lumbar, thigh and knee wearable IMU nodes during data collection

Following multi-node quality control, 30 of 32 participant/activity recordings were retained. The aligned dataset contained 81,240 observations from eight participants across sitting, standing, walking and Sit-to-Walk-labelled recordings. The Sit-to-Walk recording for user_7 and walking recording for user_8 were excluded, leaving sitting and standing available for all participants and walking and Sit-to-Walk data available for seven.

After removing Sit-to-Walk-labelled recordings from the HAR dataset, 78,430 observations remained across the 18 IMU channels. Segmentation using 100-sample windows and a 50-sample hop produced 1,536 HAR windows.

Activity	HAR Windows
Sit	512
Stand	546
Walk	478
Total	1,536

Although the overall number of windows was reasonably balanced across classes, the amount of data available for each participant was not always consistent. For example, user_2 had only four complete sitting windows, while user_8 had no

retained walking windows. This means the user_8 LOSO fold does not provide an evaluation of walking performance.

The user/activity participation heatmap below shows which activities were retained for each participant after quality control.

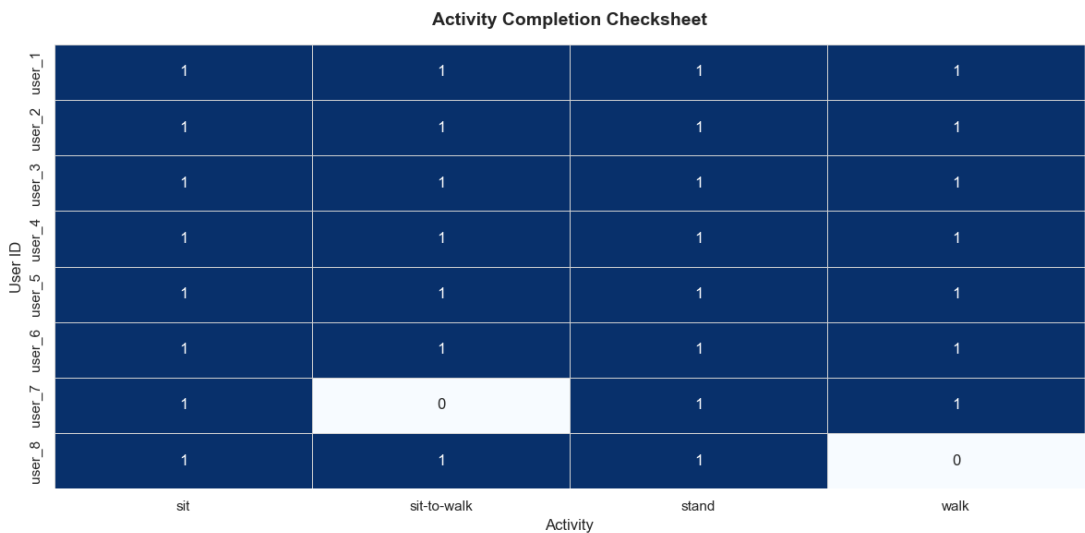


Figure 5.3 User/activity participation heatmap

In the plots 5.4 and 5.5 below, we see the distribution of samples and activities across the dataset.

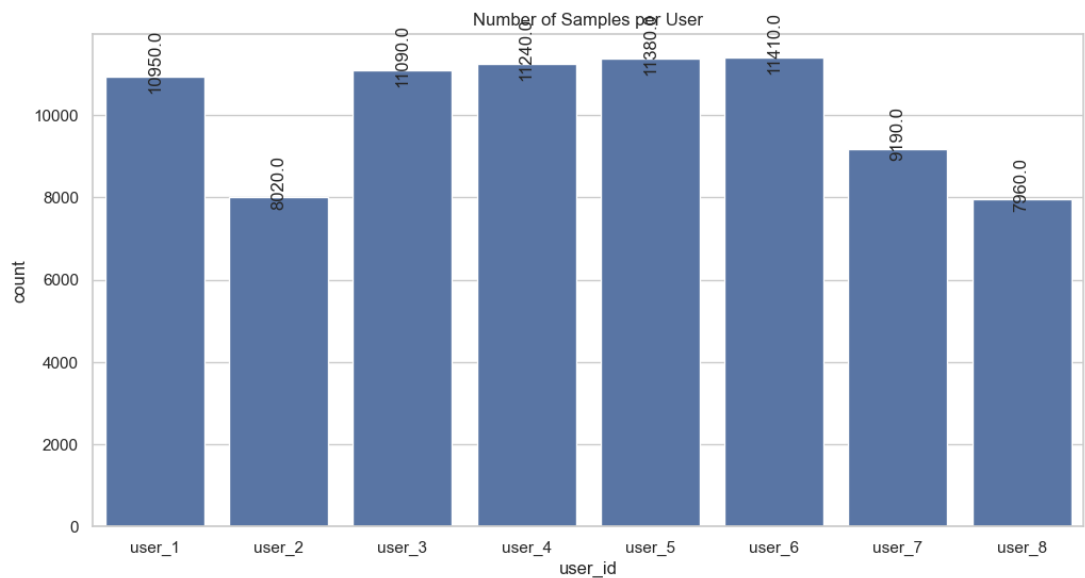


Figure 5.4: Sample counts per user.

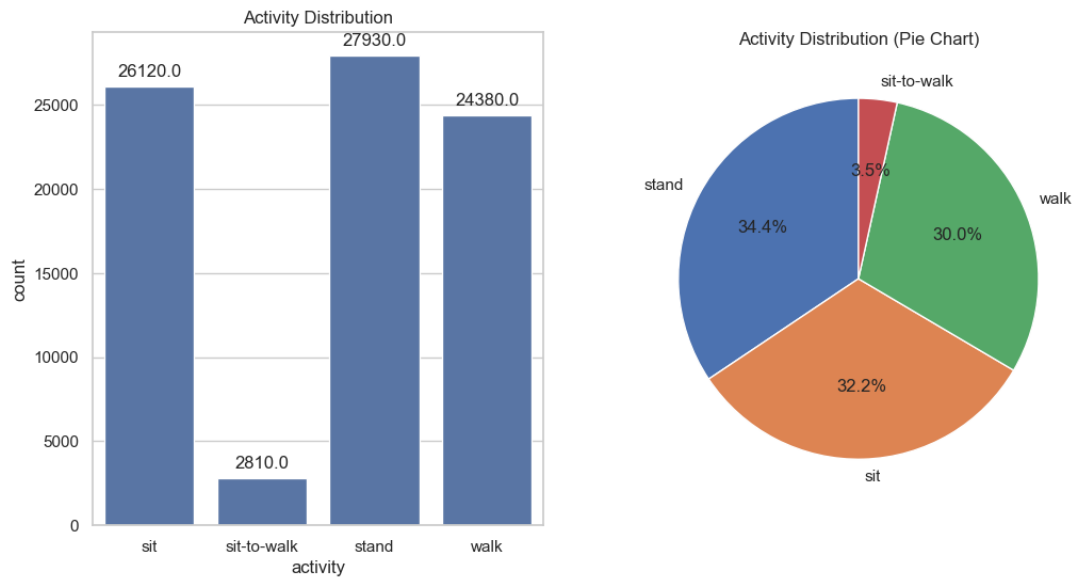
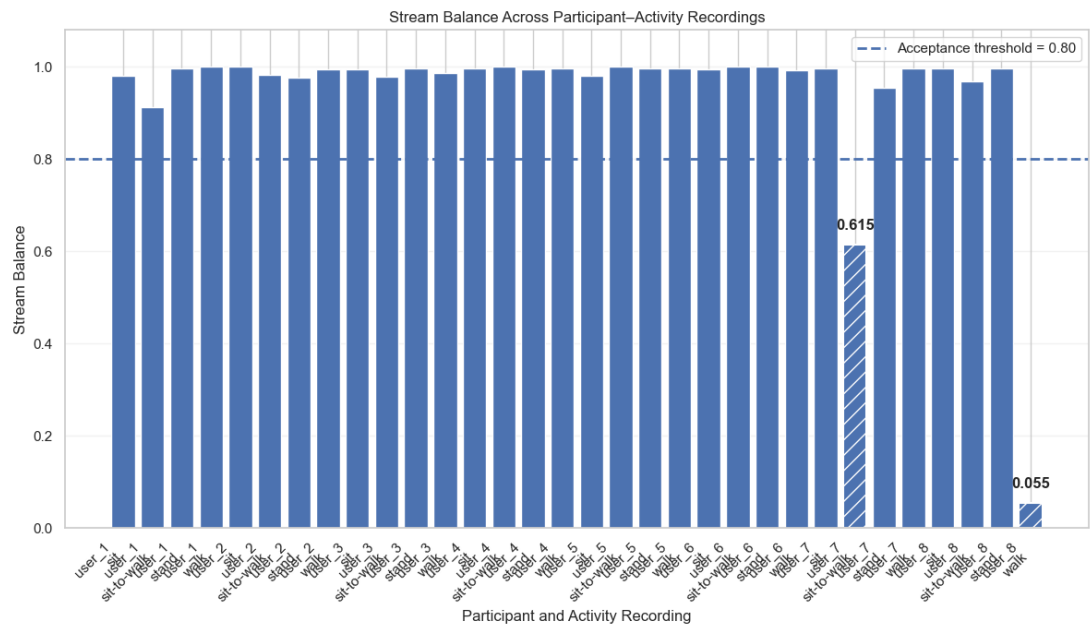


Figure 5.5: Activity distribution across the resulting dataset.

Data Quality and Signal Filtering

The two rejected recordings demonstrate the effect of the stream-quality criterion. For user_7's Sit-to-walk recording, the shortest stream contained 240 observations compared with 390 in the longest stream, giving a balance of approximately 0.615. For user_8's walking recording, only about 200 lumbar observations remained compared with approximately 3,600 knee and thigh observations, producing a balance of approximately 0.055. Both were below the required 0.80 threshold.



5.6: Stream balance across participant/activity recordings with a 0.80 acceptance threshold.

Timing checks also showed a difference between the recorded Arduino timestamps and the configured sampling interval. Many recordings had a median interval of about 17 ms, while the final firmware uses a nominal 20 ms interval. This does not prevent evaluation of fixed-length HAR windows, but time-based outputs still depend on confirming the true sampling rate.

Butterworth Lowpass Filter Visualisations

The plots below show the signal samples before and after applying the Butterworth Lowpass filter at 4Hz

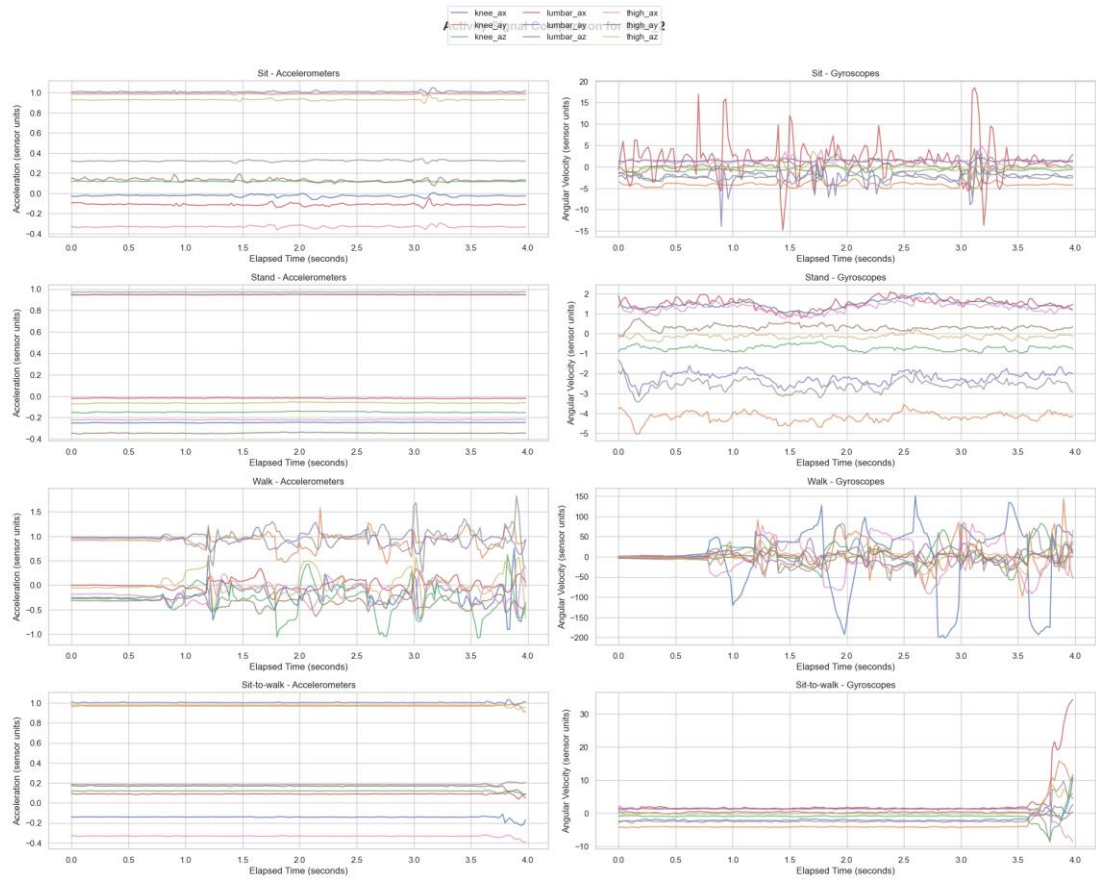


Figure 5.7: Raw accelerometer and gyroscope signals before 4 Hz low-pass filtering.

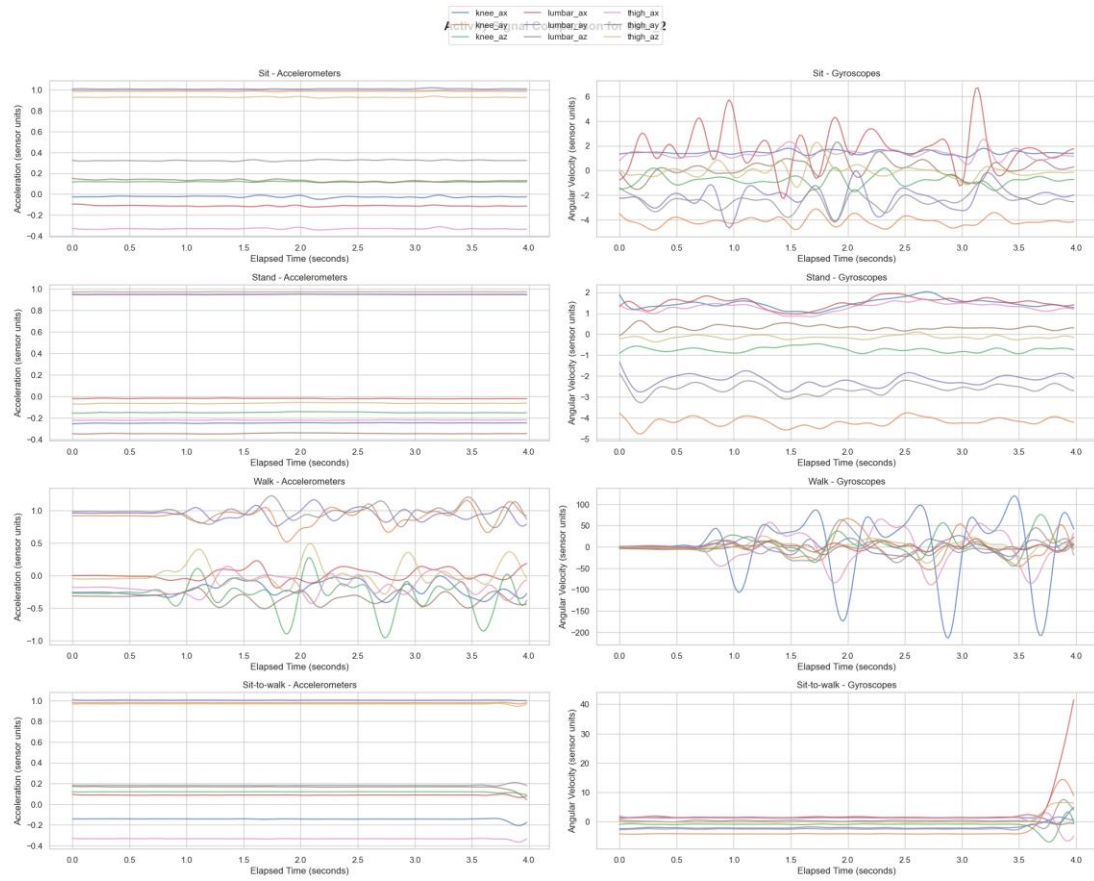


Figure 5.8: Accelerometer and gyroscope signals after 4 Hz low-pass filtering.

Participant-Independent HAR Performance

Random Forest, 1D-CNN, LSTM and Conv-LSTM were evaluated using the same outer LOSO test participants and the same underlying 18 IMU channels. Random Forest produced the strongest and most stable participant-independent performance.

Participant-average and pooled metrics differ because participant-average scores give each participant similar weight, while pooled scores are influenced more by participants who contribute more windows. Despite this difference, both approaches rank the models in the same order.

The neural models showed much greater variation between participants. For example, 1D-CNN accuracy was above 0.96 for some participants but dropped to about 0.34 for user_6 and 0.32 for user_7. Conv-LSTM showed a similar pattern, while LSTM performance also varied considerably. This suggests that the main weakness of the neural models was inconsistent performance across unseen participants, not a complete inability to recognise the activities.

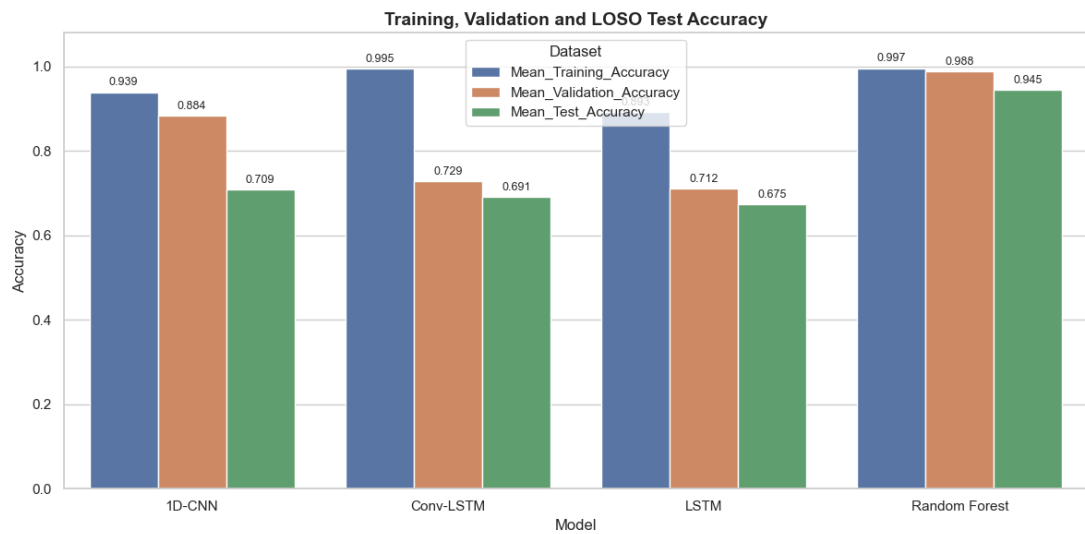


Figure 5.9: Mean training, validation and LOSO test accuracy for the four baseline model classes

Class-specific results further explain the difference between models. Random Forest achieved F1 scores of 0.9187 for sitting, 0.9250 for standing and 0.9842 for walking. In contrast, 1D-CNN achieved very high walking recall (0.9874) but low walking precision (0.5187), while LSTM standing recall fell to 0.2070. Conv-LSTM produced more balanced class recall but remained below Random Forest overall.

Random Forest Optimisation

Following the four-model comparison, Random Forest was evaluated using nested participant-level optimisation. The initial Random Forest and the nested

experiment used different development populations. The increase from the initial mean accuracy of 0.945 to approximately 0.99 therefore cannot be attributed solely to hyperparameter tuning.

The valid tuning comparison is between the baseline and tuned Random Forest models trained on the same outer-fold development participants.

The improvement attributable specifically to tuning was therefore small: approximately 0.0017 in mean accuracy and 0.0013 in weighted F1. The baseline was already close to saturation when trained on all available non-test participants.

The Figure below is a graphical representation of these findings

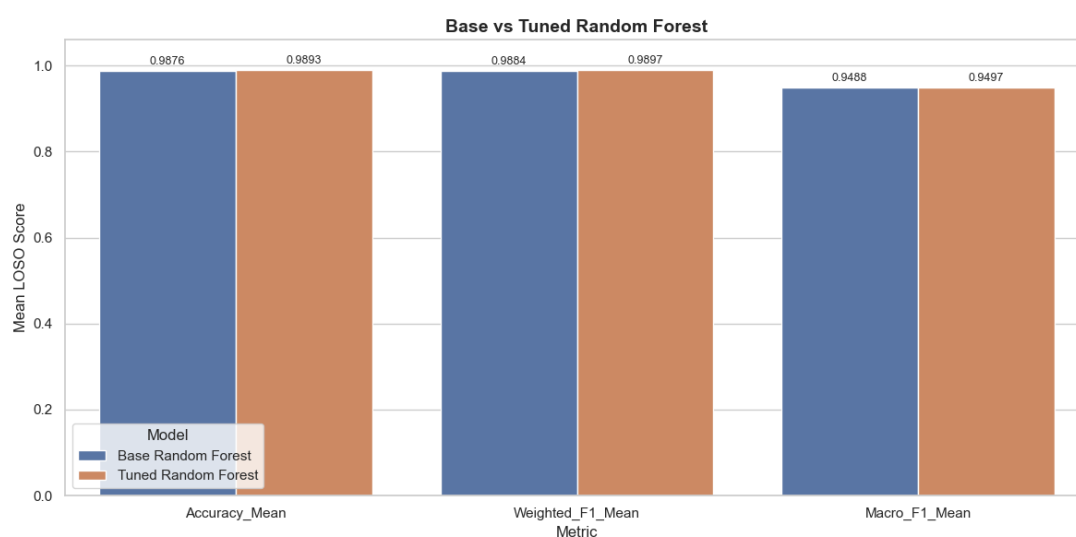


Figure 5.10: Comparison between the baseline and tuned Random Forest models.

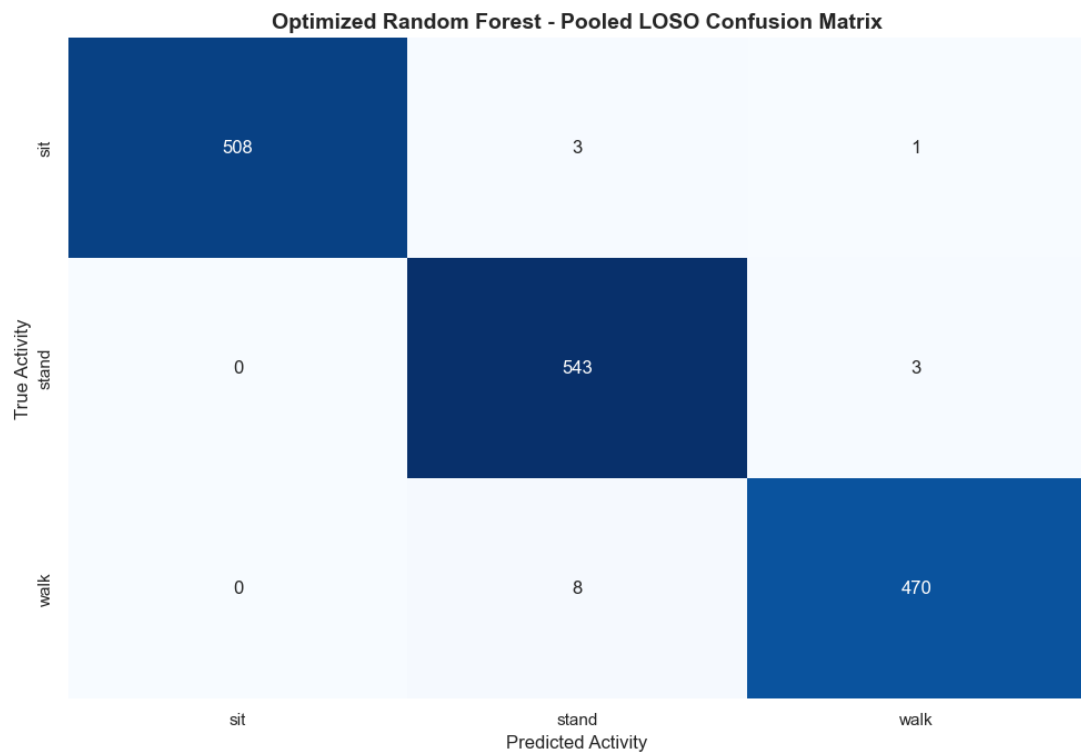


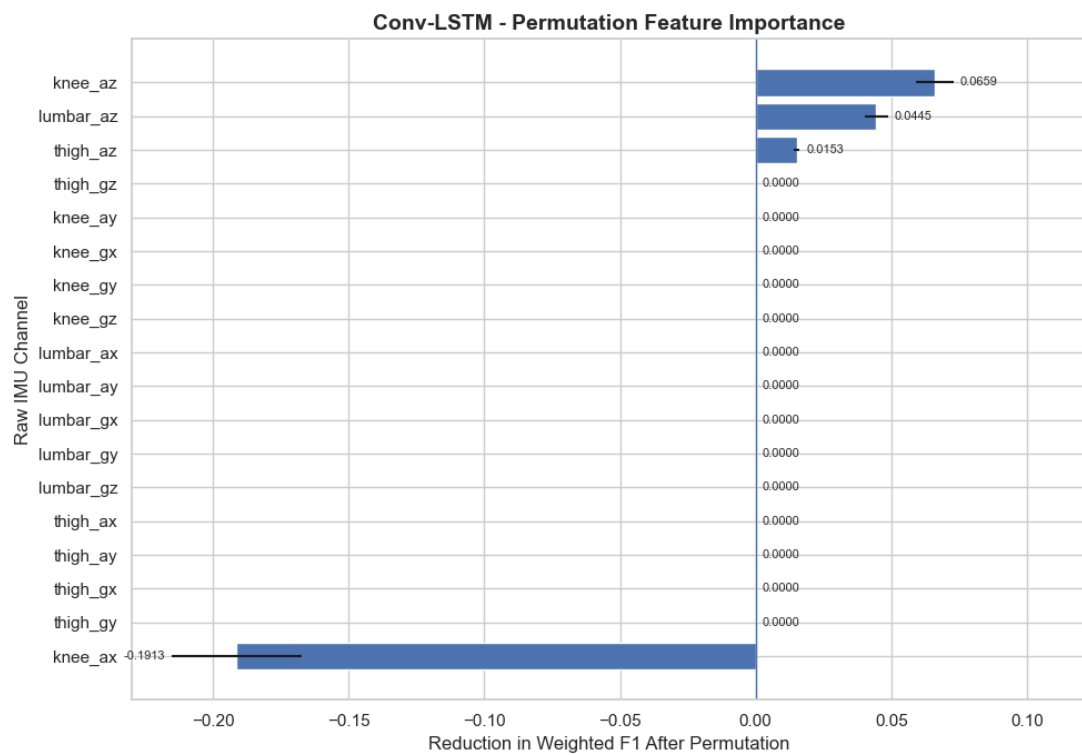
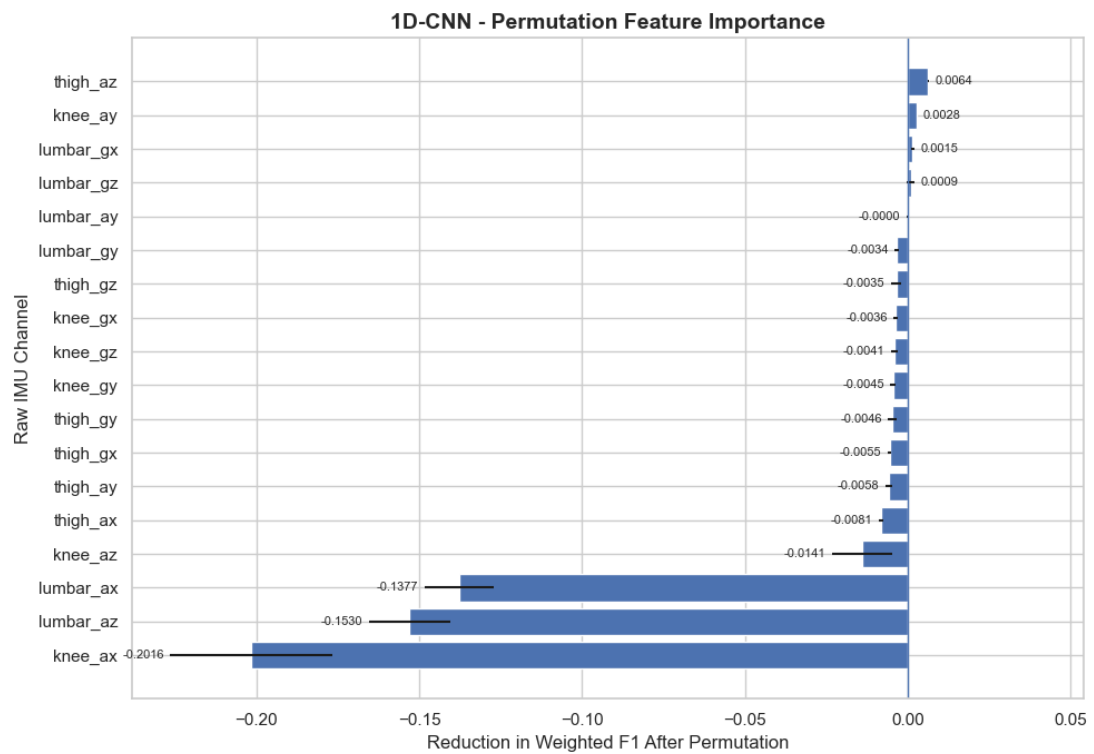
Figure 5.10: Pooled LOSO confusion matrix for the tuned Random Forest model.

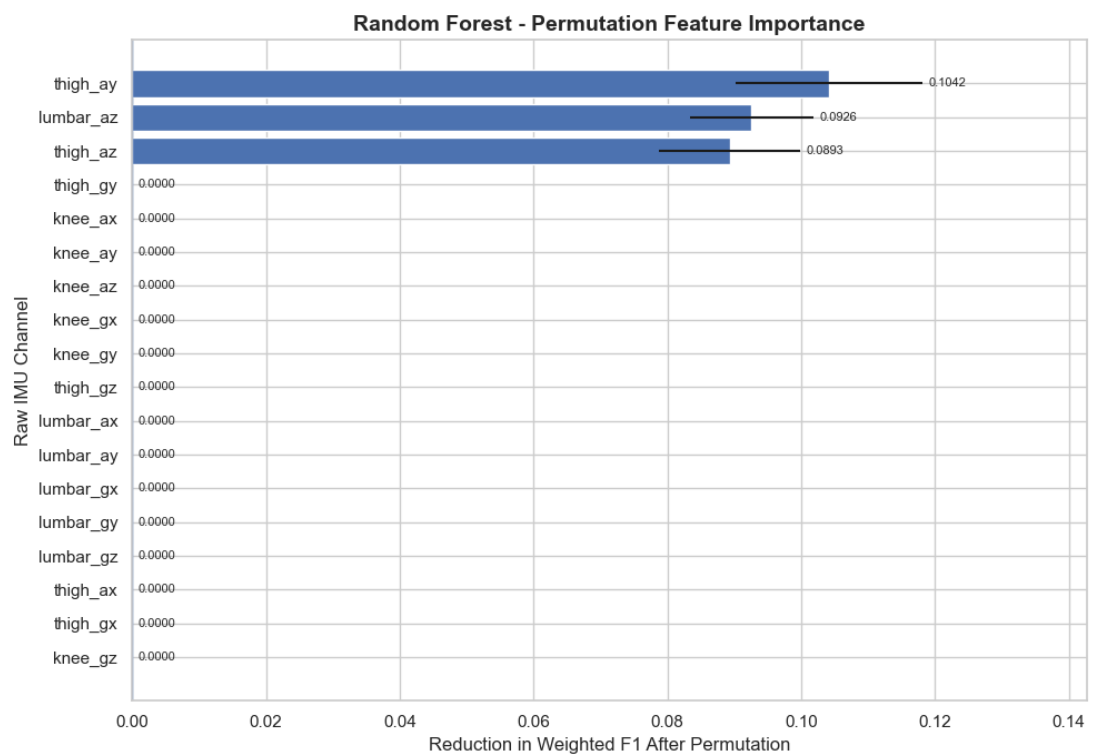
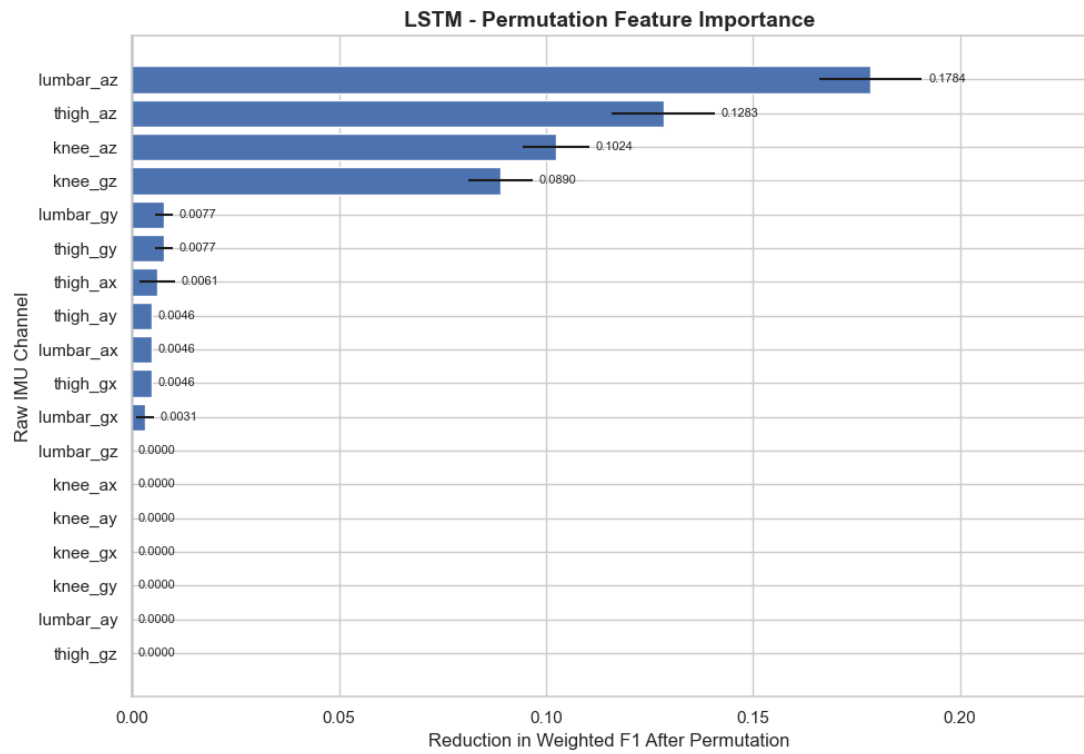
Sensor-Channel Contribution

Permutation analysis was used to examine reliance on the 18 raw IMU channels in one representative LOSO fold. The strongest channels for each model are shown in the plots below.

Lumbar and thigh acceleration channels appeared repeatedly among the most influential inputs, while Random Forest also relied strongly on thigh_ay. However, permutation values should be interpreted within each model rather than as directly comparable absolute importance scores between architectures. Correlated channels may also substitute for one another, so the results do not establish that the knee sensor is unnecessary or that the identified arrangement is universally optimal.

Figures 5.12–5.15 show the permutation importance of all 18 IMU channels for the 1D-CNN, Conv-LSTM, LSTM and Random Forest models, respectively.





TUG-Related and Walking Mobility Outputs

Seven Sit-to-Walk-labelled recordings remained after quality control and were analysed as composite TUG-related recordings. Under the configured 50 Hz analytical timeline, recording duration ranged from 6.2 to 11.4 seconds, with a mean of 8.029 seconds, median of 8.0 seconds and standard deviation of 1.867 seconds.

Walking recordings were available for seven participants. Knee-acceleration peak detection estimated 105–162 gait events per recording, with estimated cadence ranging from approximately 99.7 to 126.0 steps/min and a mean of approximately 114.8 steps/min.

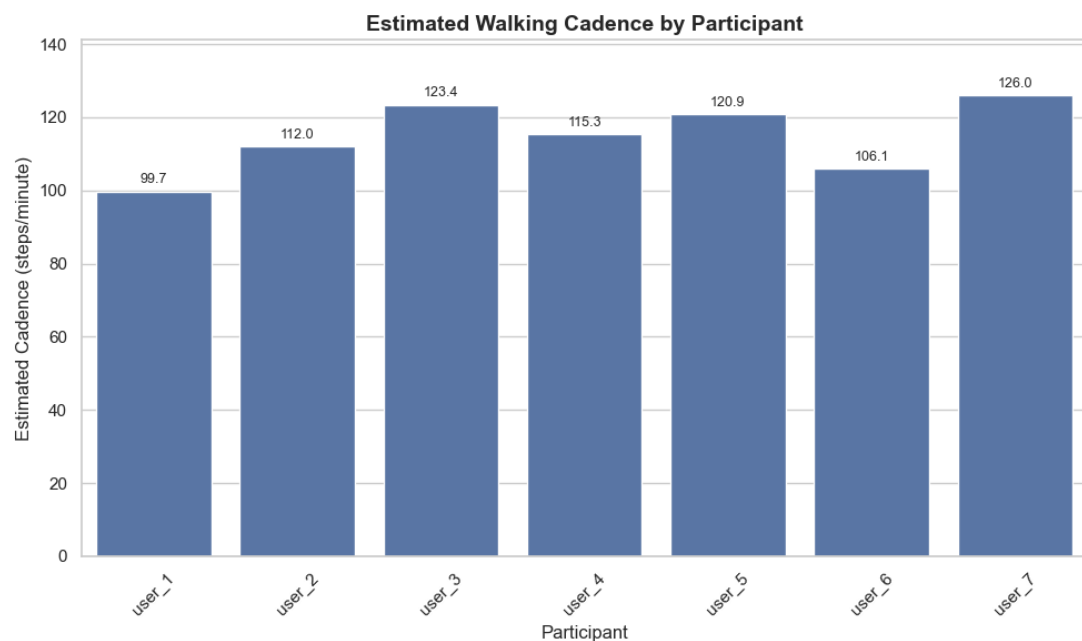


Figure 5.16: Estimated walking cadence by participant.

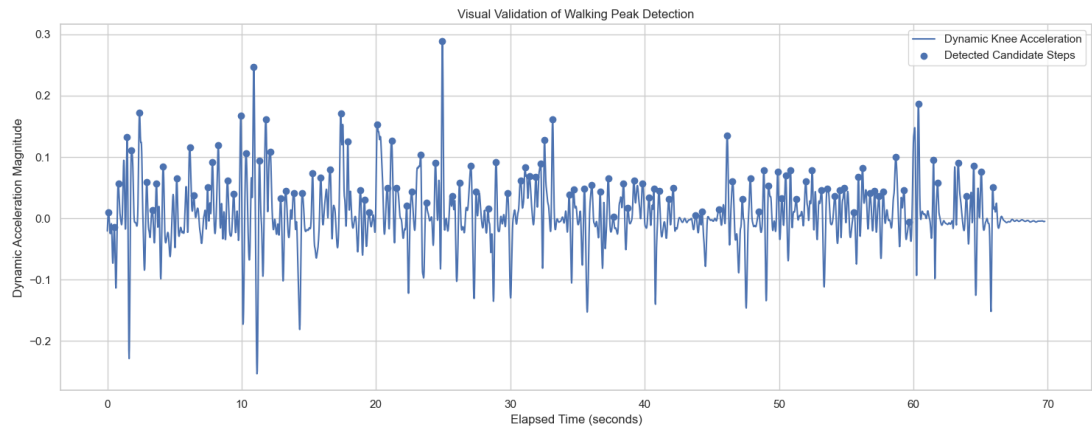


Figure 5.17: Example knee-acceleration signal with detected walking peaks.

The peak detector was not validated against manual counting, video annotation or an instrumented walkway. Detected peaks are therefore treated as estimated gait events rather than verified steps. Walking distance and speed are also not reported because participant-specific stride length was unavailable. In addition, cadence values require recalculation if the approximately 17 ms timestamp diagnostic is confirmed to represent the true physical sampling interval.

The real-time system additionally records classifier-derived changes between sit, stand and walk states. These events provide a participant-specific activity history, but their displayed duration measures the interval required to confirm a classifier state change rather than the physical duration of the movement.

Computational Performance and Real-Time Prototype

All four model families completed single-window computation within the nominal one-second interval between successive prediction windows on the benchmark system. The average single-window inference time and 95th -percentile inference latency are shown in figures 5.17 and 5.18 below

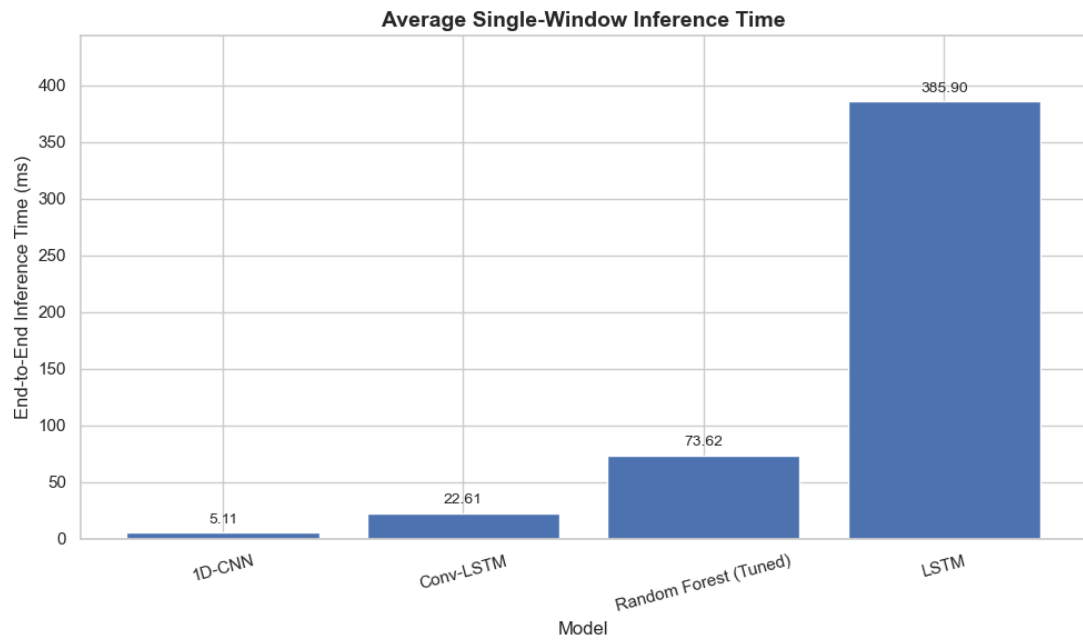


Figure 5.18: Mean single-window computational latency by model.

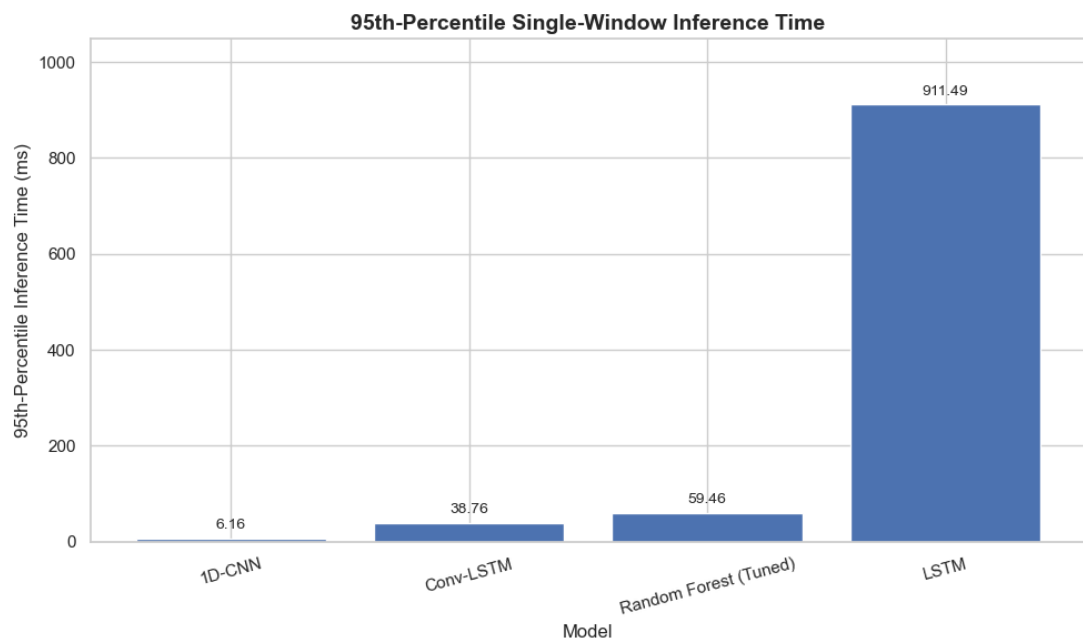


Figure 5.19: 95th-percentile computational latency by model.

The 1D-CNN was computationally fastest. Random Forest was selected because it combined substantially stronger participant-independent recognition with computational latency well below the available prediction interval.

The Random Forest benchmark includes statistical feature extraction and model prediction. For the neural architectures, the benchmark includes scaling and forward inference, with Conv-LSTM also requiring input reshaping. The neural timing experiment uses architecture-identical networks and therefore represents architecture-level computational cost rather than the latency of a specific trained LOSO fold.

Computational latency should also be distinguished from system response time. Under the nominal 50 Hz configuration, the first prediction requires approximately 100 samples, corresponding to a two-second accumulation period, while subsequent predictions occur after each 50-sample hop, nominally once per second. The prototype is therefore appropriately described as near-real-time rather than instantaneous.

The live prototype demonstrates successful integration of the wearable nodes, deployed model and browser interface. It does not constitute an independent live accuracy evaluation.

Evaluation Against the Research Questions

The results answer the research questions with different levels of evidence.

Research Question	Evaluation
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RQ1: Participant-independent HAR	The tuned Random Forest achieved mean outer-fold accuracy of approximately 98.93%, mean weighted F1 of 98.97%, and pooled accuracy of 99.02%. This supports strong recognition of sitting, standing and walking within the controlled dataset.
RQ2: Model comparison	Random Forest was substantially more stable across unseen participants than the tested 1D-CNN, LSTM and Conv-LSTM models. Increasing model complexity did not improve generalisation in this dataset.
RQ3: Sensor contribution	Lumbar and thigh acceleration channels were repeatedly influential, although one-fold permutation analysis and correlated signals prevent conclusions about an optimal reduced-sensor configuration.
RQ4: Near-real-time suitability	The selected Random Forest required approximately 73.6 ms mean computational latency and 59.5 ms at the 95th percentile, well below the nominal one-second prediction hop.
RQ5: Richer mobility representation	Activity state, estimated gait events, cadence, TUG-related recording duration and transition histories provide more information than aggregate duration alone, but exact biomechanical timing, validated step counts, gait speed,

	turning and complete standard-TUG decomposition were not established.
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Limitations

The main limitations of the study are the small number of participants and the controlled way in which the activities were recorded. Overlapping windows increased the amount of data available to the models, but they did not increase the number of independent participants. Some activities were also unavailable for certain participants after quality control. In addition, the three sensor streams were only approximately aligned, and uncertainty around the true sampling interval affects measurements that depend on physical time.

The mobility outputs also have different levels of validation. Step detection was not compared with an external reference. Turning was not collected, and transition duration is based on classifier state changes, and not direct biomechanical measurement. The live prototype was not evaluated using a separate annotated accuracy test, and the filtering used during live inference is not identical to the offline filtering process.

The study also did not include prospective fall outcomes or longitudinal rehabilitation data. The findings therefore support activity recognition on unseen participants and proof-of-concept mobility analysis, but they do not demonstrate validated fall prediction or clinical rehabilitation effectiveness.

Chapter Summary

This chapter evaluated the multi-node wearable system across dataset quality, participant-independent HAR, sensor contribution, exploratory mobility analysis and computational performance. Random Forest produced the strongest and most stable unseen-participant classification, with nested optimisation yielding approximately 98.93% mean LOSO accuracy and 99.02% pooled accuracy.

Lumbar and thigh acceleration channels were influential across several models, while TUG-related duration, estimated gait events, cadence and classifier-derived transitions demonstrated additional participant-specific mobility outputs. For all candidate architectures, the reported mean and 95th-percentile single-window computational latencies remained below the nominal one-second prediction interval, supporting near-real-time feasibility on the benchmark system. These findings demonstrate technical feasibility within the available controlled dataset while retaining clear limits around clinical interpretation and external validation.

CHAPTER 6

CONCLUSIONS / FUTURE WORK

Introduction

This dissertation investigated whether a three-node wearable IMU system could recognise sitting, standing and walking in unseen participants and use these recognised movement states to provide additional gait and TUG-related mobility information.

The study combined wearable sensing, signal processing, Human Activity Recognition, sensor analysis and near-real-time deployment. This chapter summarises the main findings, discusses the limitations of the study and identifies areas for future development.

Summary of Findings

The developed system used IMU sensors positioned at the knee, thigh and lumbar regions to collect acceleration and angular-velocity data during sitting, standing, walking and Sit-to-Walk-labelled (TUG-related recordings). After quality control, the sensor streams were approximately aligned, filtered and segmented before Random Forest, 1D-CNN, LSTM and Conv-LSTM models were evaluated using Leave-One-Subject-Out testing.

Random Forest produced the strongest and most consistent performance on unseen participants. After participant-level optimisation, the model achieved approximately 99% pooled LOSO accuracy. The neural models performed well for some participants but showed greater variation between participants,

suggesting that the amount and diversity of training data may have limited their ability to generalise consistently.

The study also showed that the wearable system could provide more information than activity labels alone. The prototype produced estimated step count, estimated cadence, TUG-related recording duration and classifier-derived transitions between sitting, standing and walking. These outputs provide additional information about movement, although they were not all validated against independent reference measurements.

Sensor-channel analysis showed that lumbar and thigh acceleration signals were influential across several models in the representative analysis. However, the rankings varied between architectures, so this does not show that any of the three sensors can be removed without affecting performance. A complete sensor-ablation experiment would be needed to answer this.

The selected Random Forest was also fast enough for near-real-time use and was integrated into the live monitoring application. This demonstrated that the wearable sensors, processing pipeline, activity classifier and mobility outputs could operate together as one system.

Overall, the study achieved its main aim at the level of a technical prototype. It demonstrated activity recognition on unseen participants, provided selected mobility information and supported near-real-time monitoring. It did not evaluate rehabilitation improvement or directly predict fall risk.

Main Contribution

The main contribution of this project is the integration of multi-node wearable sensing, activity recognition, mobility analysis and near-real-time monitoring within one system.

An important part of the study was the use of participant-level evaluation. Windows belonging to the held-out participant were excluded from model development and used only for testing, providing a more realistic evaluation of performance on unseen individuals.

The study also avoided producing measurements that were not supported by the collected data. Walking distance and gait speed were not calculated because stride length or distance was not measured. Turning was not analysed because dedicated turning data were not collected, and classifier-derived transitions were not treated as direct biomechanical transition measurements.

Limitations

The main limitation of the study is the small participant sample. Eight participants were sufficient for an initial technical evaluation but are not enough to demonstrate generalisation across wider rehabilitation or clinical populations. The activities were also collected under controlled conditions, so performance during free-living movement remains unknown.

The three wearable nodes did not share the same hardware clock, meaning that sensor alignment was approximate. The difference between the observed Arduino timestamp interval of approximately 17 ms and the configured 20 ms sampling interval also creates uncertainty for measurements that depend directly on physical time.

There were no complete standard TUG recordings because dedicated turning data and phase-level annotations were not collected.

Step count and cadence remain estimates because they were not compared with an independent reference system. The live system was also not evaluated using a separate labelled real-time accuracy experiment.

Finally, the study did not include longitudinal rehabilitation data or fall outcomes. The findings therefore support wearable activity recognition and proof-of-concept mobility analysis, but they do not demonstrate rehabilitation effectiveness or validated fall-risk prediction.

Future Work

Future work should first focus on collecting a larger and more diverse dataset. Increasing the amount of training data would make it possible to investigate whether the 1D-CNN, LSTM and Conv-LSTM models improve when more examples are available. The neural models showed strong performance for some participants but were less consistent across unseen participants, so a larger dataset would help determine whether this was mainly caused by limited training data.

Future data collection should also include more complete TUG recordings, including the 180° turn and clearly identified movement phases. Reference measurements such as video annotations or manually counted steps could be used to validate transition timing and step detection. A known walking distance would also allow gait speed to be calculated without assuming stride length.

Another important extension would be to collect TUG and mobility data from individuals who have experienced falls and those who have not. With a larger labelled dataset, a separate model could then be developed to investigate whether TUG-related IMU information can classify fall risk. This would be a different prediction task from the current HAR system and would require appropriate fall-related labels and independent validation.

Future hardware development should also improve synchronisation between the three sensor nodes. This could be achieved using shared timing, sequence numbers or improved clock synchronisation. A complete sensor-ablation study

could then compare one-sensor, two-sensor and three-sensor configurations to determine whether the same recognition performance can be achieved with fewer wearable nodes.

For rehabilitation monitoring, future studies should collect repeated measurements from the same participants over time. This would make it possible to investigate whether changes in cadence, walking performance, transitions or other validated mobility information reflect rehabilitation progress.

Final Conclusion

This project developed and evaluated a three-node wearable IMU system for recognising sitting, standing and walking and deriving selected mobility information. Random Forest provided the strongest and most consistent performance for the available dataset and was successfully integrated into a near-real-time monitoring application.

The study shows that the wearable system can provide more detailed mobility information than a single activity label or overall task duration. However, estimated step count, cadence, TUG-related duration and activity transitions still have different levels of validation.

The resulting system should therefore be viewed as a technical proof of concept for wearable mobility assessment. Future work should focus on collecting more data, improving sensor synchronisation, validating mobility measurements, expanding TUG data collection and investigating whether larger datasets can improve neural-model performance and support a separate fall-risk classification model.

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APPENDIX A